

DATA VISUALISATION

Mini Workshop - Introduction & Tips

Pham Tan Phat

tanphat.leo@gmail.com

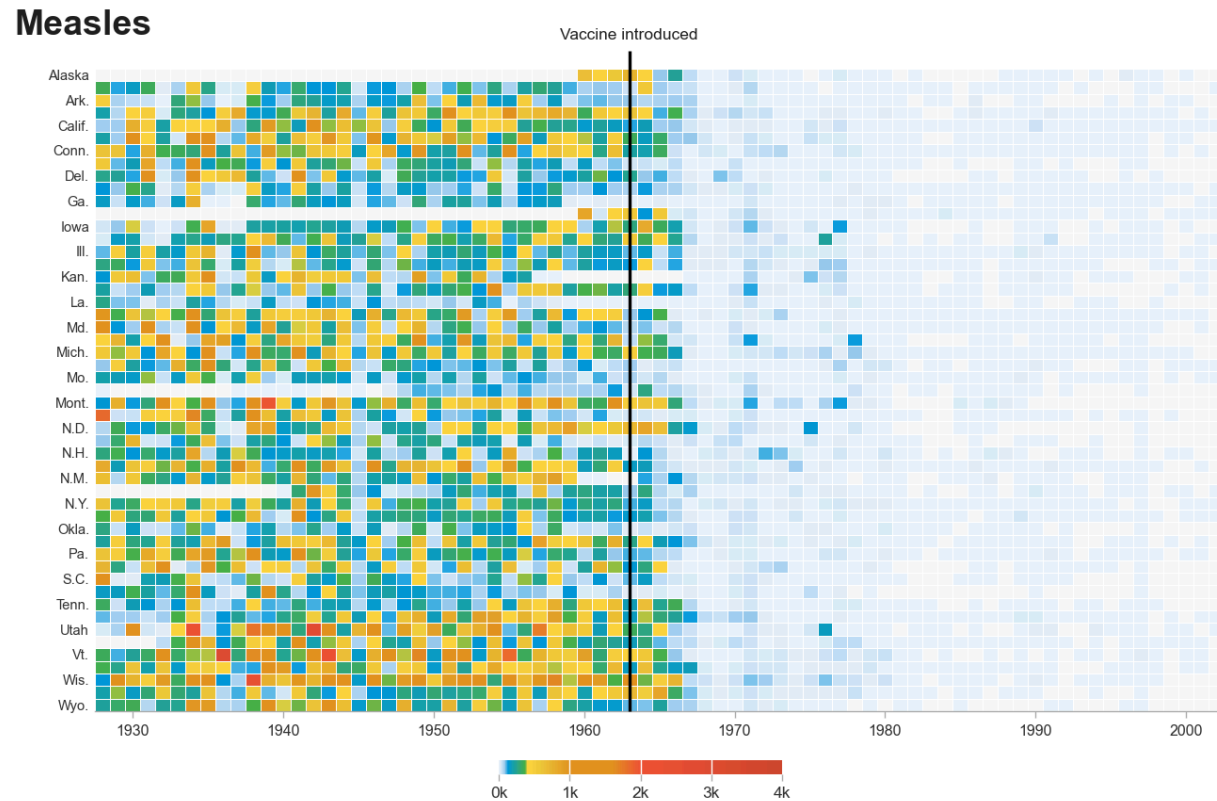
WHY VISUALIZATION?

What insight can you **quickly** draw from the data?

disease	state	year	weeks_reporting	count	population	per_htp
Hepatitis A	Alabama	1966	50	321	3,345,787	9.59
Hepatitis A	Alabama	1967	49	291	3,364,130	8.65
Hepatitis A	Alabama	1968	52	314	3,386,068	9.27
Hepatitis A	Alabama	1969	49	380	3,412,450	11.14
Hepatitis A	Alabama	1970	51	413	3,444,165	11.99
Hepatitis A	Alabama	1971	51	378	3,481,798	10.86
Hepatitis A	Alabama	1972	45	342	3,524,543	9.70
Hepatitis A	Alabama	1973	45	467	3,571,209	13.08
Hepatitis A	Alabama	1974	45	244	3,620,548	6.74
Hepatitis A	Alabama	1975	46	286	3,671,246	7.79
Hepatitis A	Alabama	1976	50	220	3,721,914	5.91
Hepatitis A	Alabama	1977	43	206	3,771,085	5.46
Hepatitis A	Alabama	1978	41	203	3,817,217	5.32
Hepatitis A	Alabama	1979	47	257	3,858,703	6.66
Hepatitis A	Alabama	1980	37	200	3,893,888	5.14
...	...	...	...	...	...	...

WHY VISUALIZATION?

What insight can you **quickly** draw from the data?



Note: cases per 100,000 people, from weekly reports to the CDC. State-years where no weeks were reported, and the years before Alaska and Hawaii were states, are left blank rather than drawn as zero.

WHY VISUALIZATION?

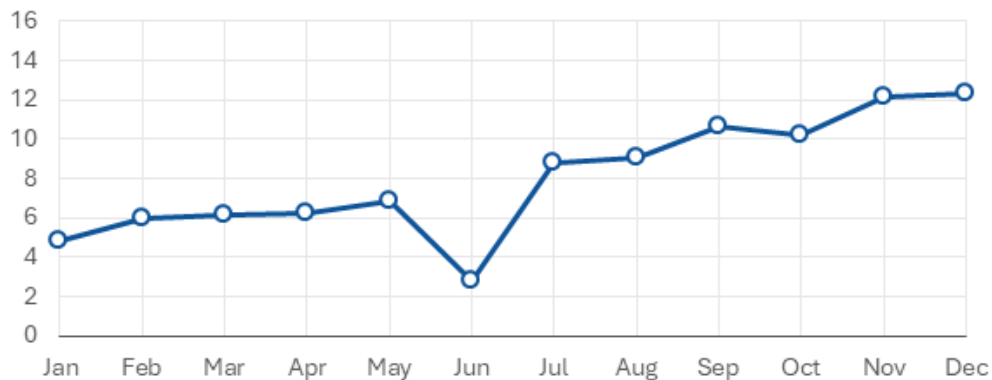
What insight can you **quickly** draw from the data?

Month	Product A	Product B	Product C	Product D
Jan	4.81	4.09	6.18	8.96
Feb	5.95	4.28	6.27	8.18
Mar	6.18	4.71	6.38	13.40
Apr	6.27	5.38	6.51	7.91
May	6.82	6.23	6.71	6.74
Jun	2.78	7.42	6.81	12.35
Jul	8.75	8.71	6.91	6.48
Aug	9.09	9.89	7.05	5.95
Sep	10.66	10.77	7.12	10.14
Oct	10.21	11.29	7.32	4.24
Nov	12.17	11.56	13.99	3.21
Dec	12.31	11.67	14.75	8.44

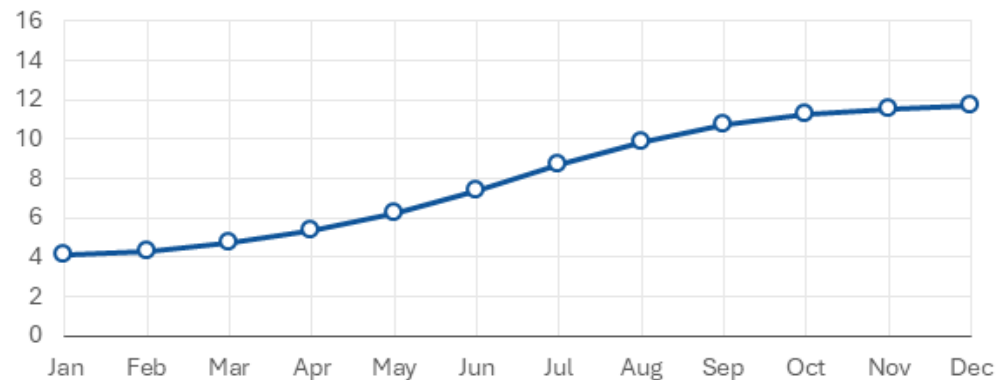
Statistic	Product A	Product B	Product C	Product D
n	12	12	12	12
Mean	8.00	8.00	8.00	8.00
Std deviation	3.00	3.00	3.00	3.00
Sample variance	9.00	9.00	9.00	9.00
Median	7.79	8.07	6.86	8.05
Minimum	2.78	4.09	6.18	3.21
Lower quartile	6.12	5.21	6.48	6.35
Upper quartile	10.32	10.90	7.17	9.26
Maximum	12.31	11.67	14.75	13.40
Range	9.53	7.58	8.57	10.19

WHY VISUALIZATION?

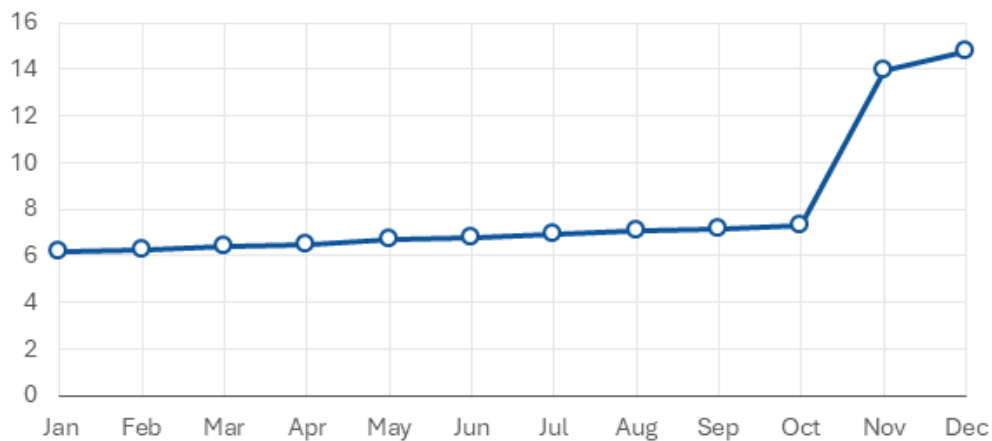
Product A is growing, one bad month aside



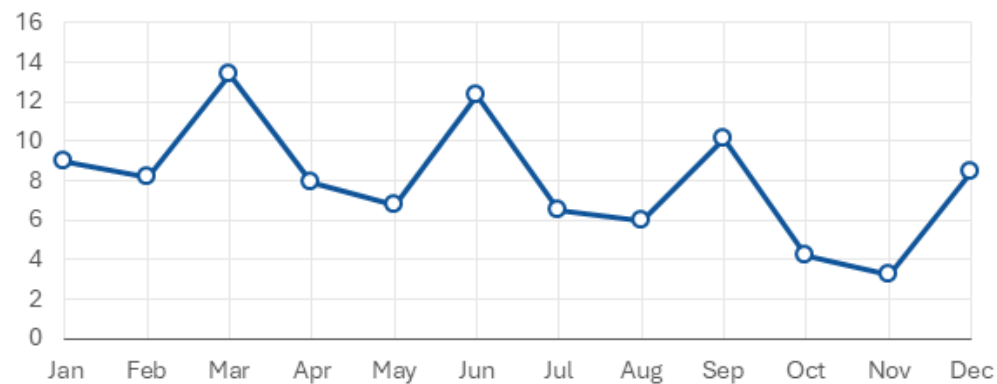
Product B's growth has already flattened



Product C did nothing until November



Product D is declining, but does well at every quarter end



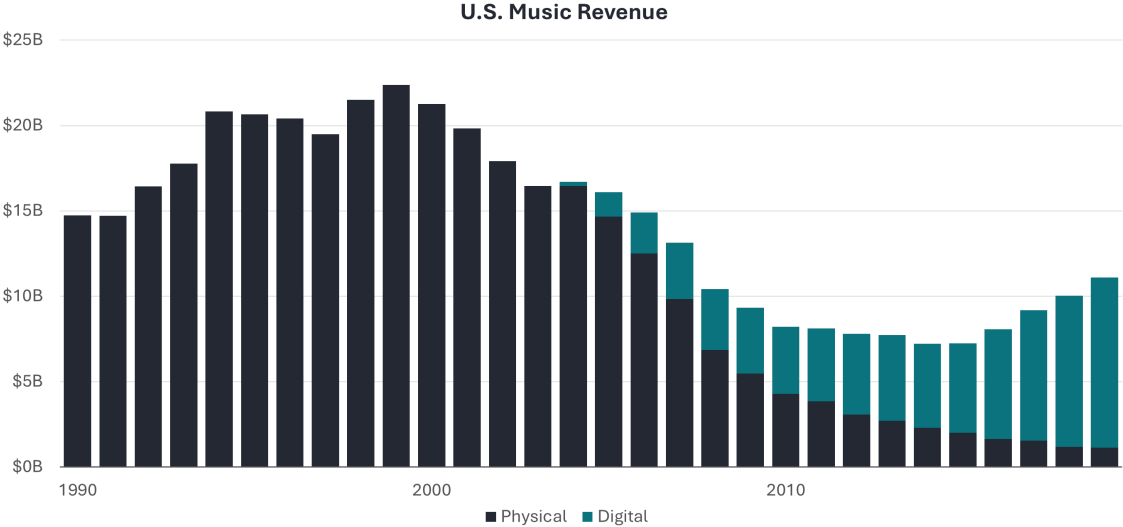
WHY VISUALIZATION?

EFFECTIVE COMMUNICATION

DATA VISUALIZATION & STORY TELLING

RAW DATA

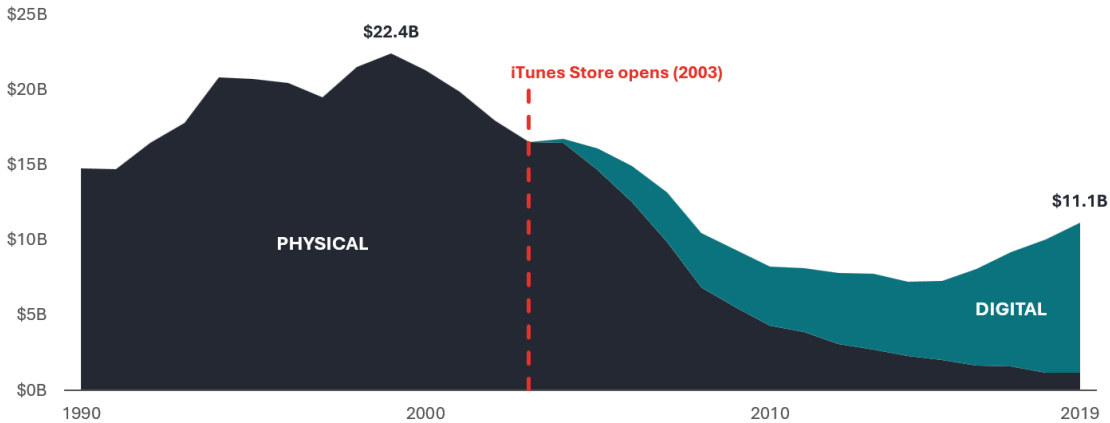
REVENUE (\$MM)			
Year	Physical	Digital	Total
1990	14,751	0	14,751
1991	14,705	0	14,705
1992	16,444	0	16,444
1993	17,775	0	17,775
1994	20,818	0	20,818
1995	20,668	0	20,668
1996	20,423	0	20,423
1997	19,492	0	19,492
1998	21,505	0	21,505
1999	22,381	0	22,381
2000	21,266	0	21,266
2001	19,836	0	19,836
2002	17,926	0	17,926
2003	16,471	0	16,471
2004	16,450	258	16,708
2005	14,655	1,433	16,088
2006	12,515	2,398	14,913
2007	9,849	3,287	13,136
2008	6,848	3,574	10,422
2009	5,484	3,848	9,332
2010	4,295	3,928	8,223
2011	3,843	4,267	8,110
2012	3,087	4,725	7,812
2013	2,704	5,016	7,720
2014	2,291	4,938	7,229
2015	2,009	5,230	7,239
2016	1,654	6,419	8,073
2017	1,560	7,615	9,175
2018	1,176	8,849	10,025
2019	1,148	9,963	11,111



DATA
VISUALIZATION

Digital revenue climbed steadily with more internet adoption

However, the music industry **shrunk 50%** comparing to 1999



DATA
STORYTELLING

SCIENCE OF VISUAL PROCESSING

The human brain **CRAVES** for:

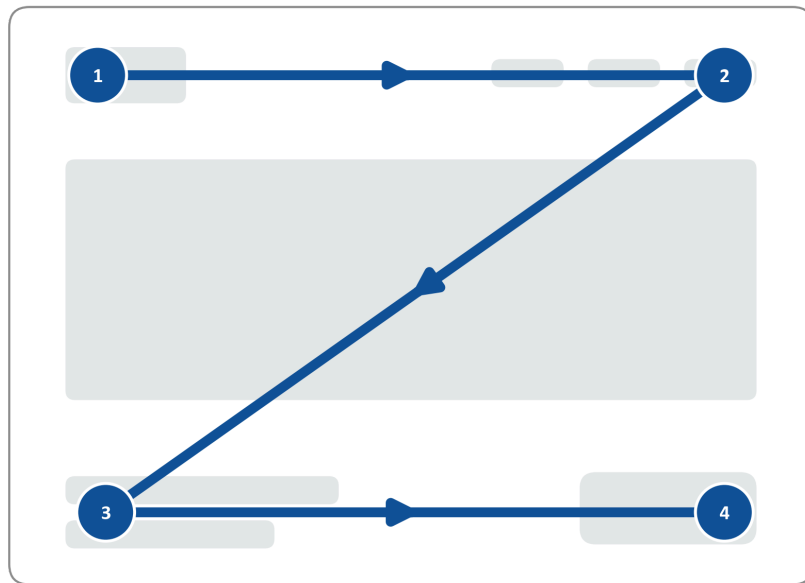
PATTERNS



VISUAL CUES

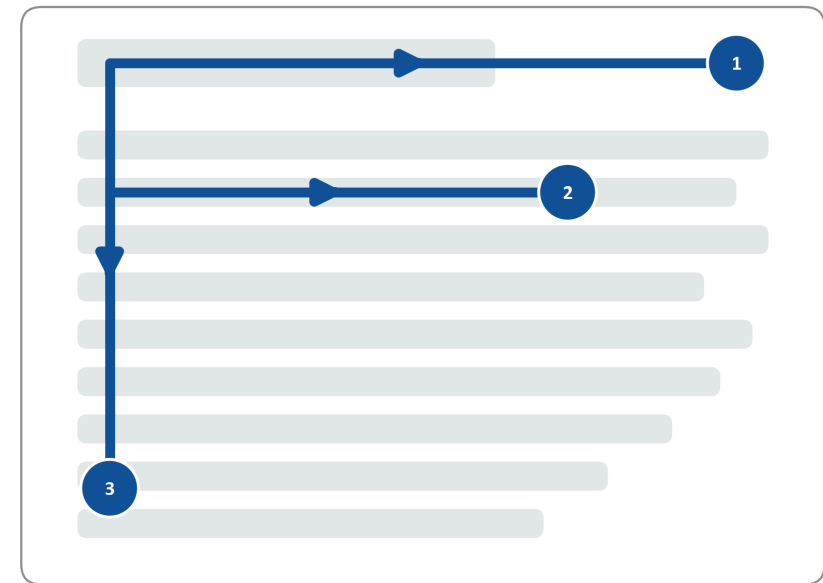
SCIENCE OF VISUAL PROCESSING

Nobody reads a page. They scan it — and the scanning route is predictable.



Z-PATTERN

Sparse pages — landing pages, ads, slides

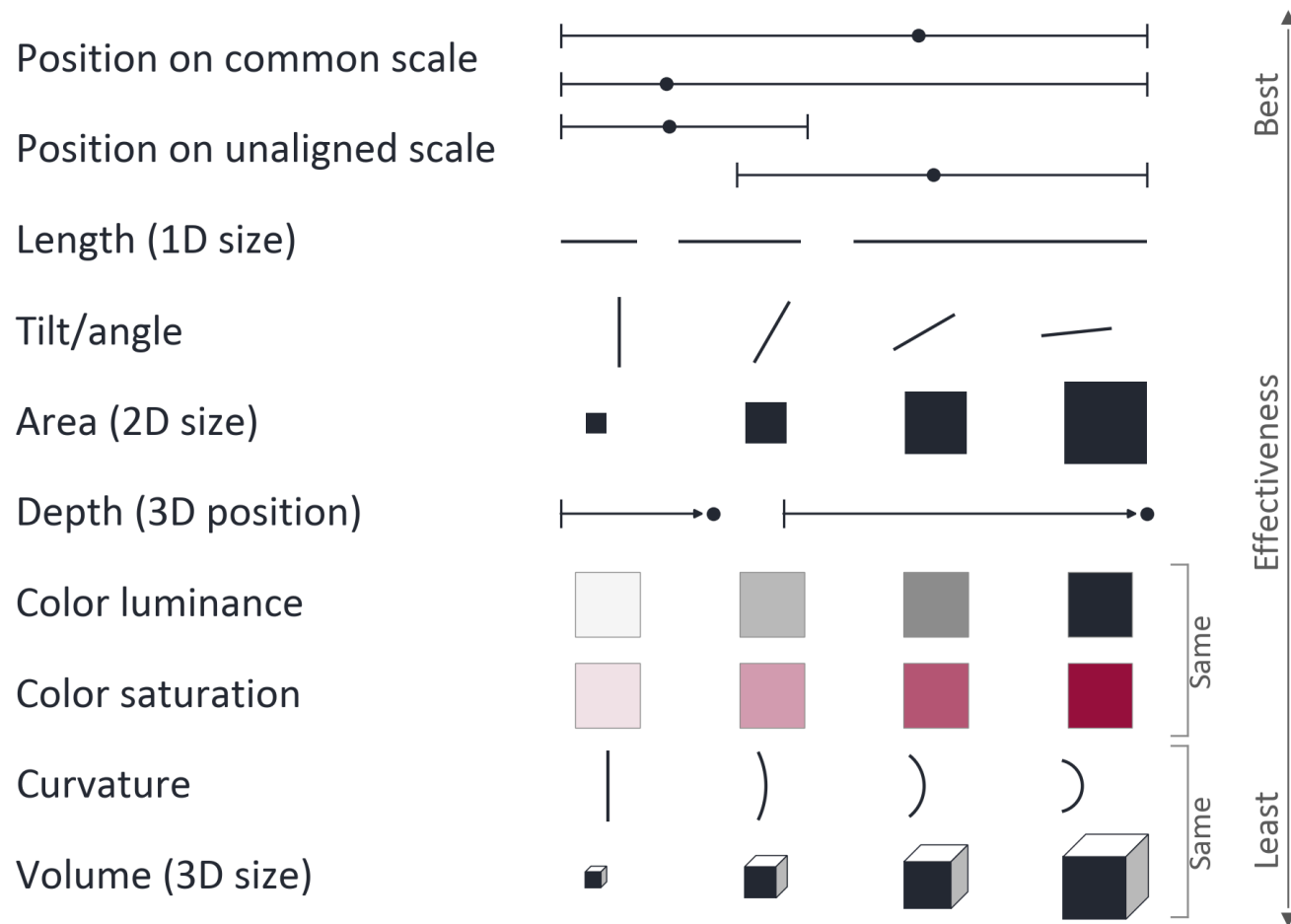


F-PATTERN

Dense pages — articles, search results, reports

Z-pattern: design convention, no originating study — after the Gutenberg diagram, E. C. Arnold (1950s). F-pattern: Nielsen, *F-Shaped Pattern for Reading Web Content*, Nielsen Norman Group (2006); revisited by Pernice (2017).

SCIENCE OF VISUAL PROCESSING



Cues (channels) that answer **MAGNITUDE** — and we read some far more accurately than others.

After Munzner, Visualization Analysis and Design (2014); ranking after Cleveland & McGill (1984)

SCIENCE OF VISUAL PROCESSING

Cues (channels) that answer **IDENTITY**

Spatial region



Color hue



Motion

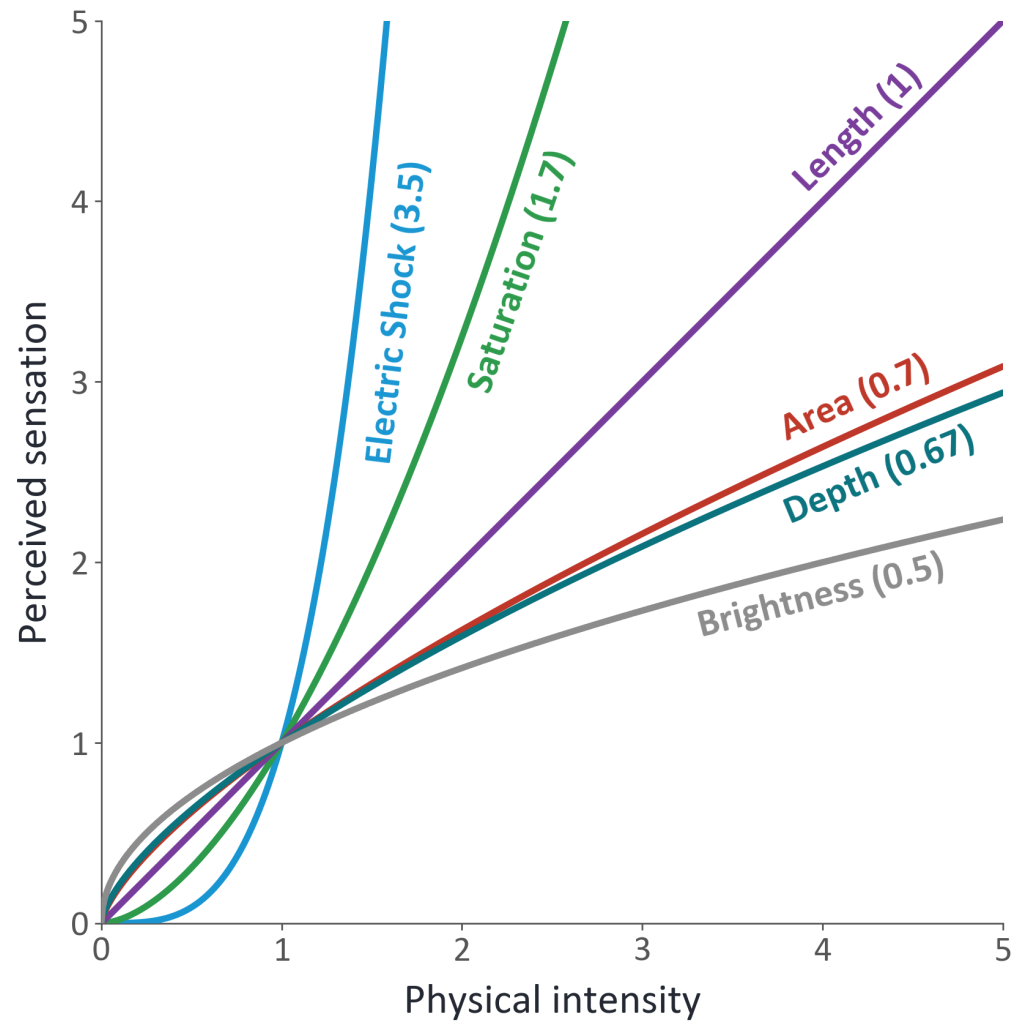


Shape



After Munzner, Visualization Analysis and Design (2014);
ranking after Cleveland & McGill (1984)

SCIENCE OF VISUAL PROCESSING



Exponents after Stevens, "On the psychophysical law", Psychological Review 64(3), 1957.
Depth (0.67) is not in Stevens's table — it is carried from the visualization literature.

length is read back faithfully.

KEY QUESTIONS WHEN VISUALIZING

1 Choose the right chart — what is the data type?

Continuous · Discrete · Nominal · Ordinal · Temporal · Geospatial

2 Choose the right chart — what does your data need to say?

ABELA Comparison · Distribution · Relationship · Composition

FT VISUAL VOCABULARY Deviation · Correlation · Change over time · Ranking · Distribution · Part of whole · Magnitude · Spatial · Flow

3 Who is the audience?

Analyst · Manager · Executive

1. WHAT IS THE DATA TYPE?

QUANTITATIVE — NUMBERS YOU MEASURE OR COUNT

Continuous

Any value in a range — between any two points there is always another.

E.G. Revenue \$47,382.56 · Temperature 36.8°C · Margin 12.4%

CHARTS Line · Scatter · Histogram · Box plot

Discrete

Countable whole numbers. There is no such thing as 2.5 customers.

E.G. Customers 847 · Units sold 1,200 · Tickets 7

CHARTS Bar · Column · Dot plot

CATEGORICAL — LABELS YOU GROUP BY

Nominal

No natural order — North is not greater than South.

E.G. Region · Product · Channel · Country

CHARTS Bar · Donut · Treemap

Ordinal

Ordered — but the gap between levels is not equal.

E.G. Poor → Excellent · Low → Critical · NPS bands

CHARTS Sorted bar · Heatmap · Stacked bar

1. WHAT IS THE DATA TYPE?

TEMPORAL — POINTS IN TIME, OR INTERVALS

Regular intervals

Daily prices, monthly sales, hourly readings. The gap is constant, so trend and seasonality are visible.

CHARTS Line · Area · Candlestick · Calendar heatmap

Irregular intervals

Purchase timestamps, log events, support tickets. Unevenly spaced — aggregate first (hourly → daily → monthly).

CHARTS Line after aggregation · Event timeline

GEOSPATIAL — ANYTHING WITH A LOCATION

Named regions

Country, province, postal code. Values join to boundaries, so the map shows a total per area.

CHARTS Choropleth

Exact coordinates

Latitude and longitude. Every row is one point, so the map shows position and density, not totals.

CHARTS Dot / bubble map · Geo heatmap

Golden rule: time goes on the X axis, the metric goes on the Y axis.

1. WHAT IS THE DATA TYPE?

Why DATA TYPES matters?

DATA TYPE	EXAMPLES	SUITABLE CHARTS
Continuous	Revenue, temperature, height	Line, scatter, histogram
Discrete	Customers, units, page views	Bar, dot plot
Nominal	Country, product name, channel	Bar, treemap
Ordinal	Rating: Poor / Good / Excellent	Sorted bar, heatmap
Temporal	Dates, timestamps	Line, area, candlestick
Geospatial	City, coordinates, country	Choropleth, dot map

2. WHAT DOES YOUR DATA NEED TO SAY?

TWO FRAMEWORKS FOR CHOOSING A CHART.

Abela's Decision Tree

Start from what you want to show — comparison, distribution, relationship, composition — then follow the branches.

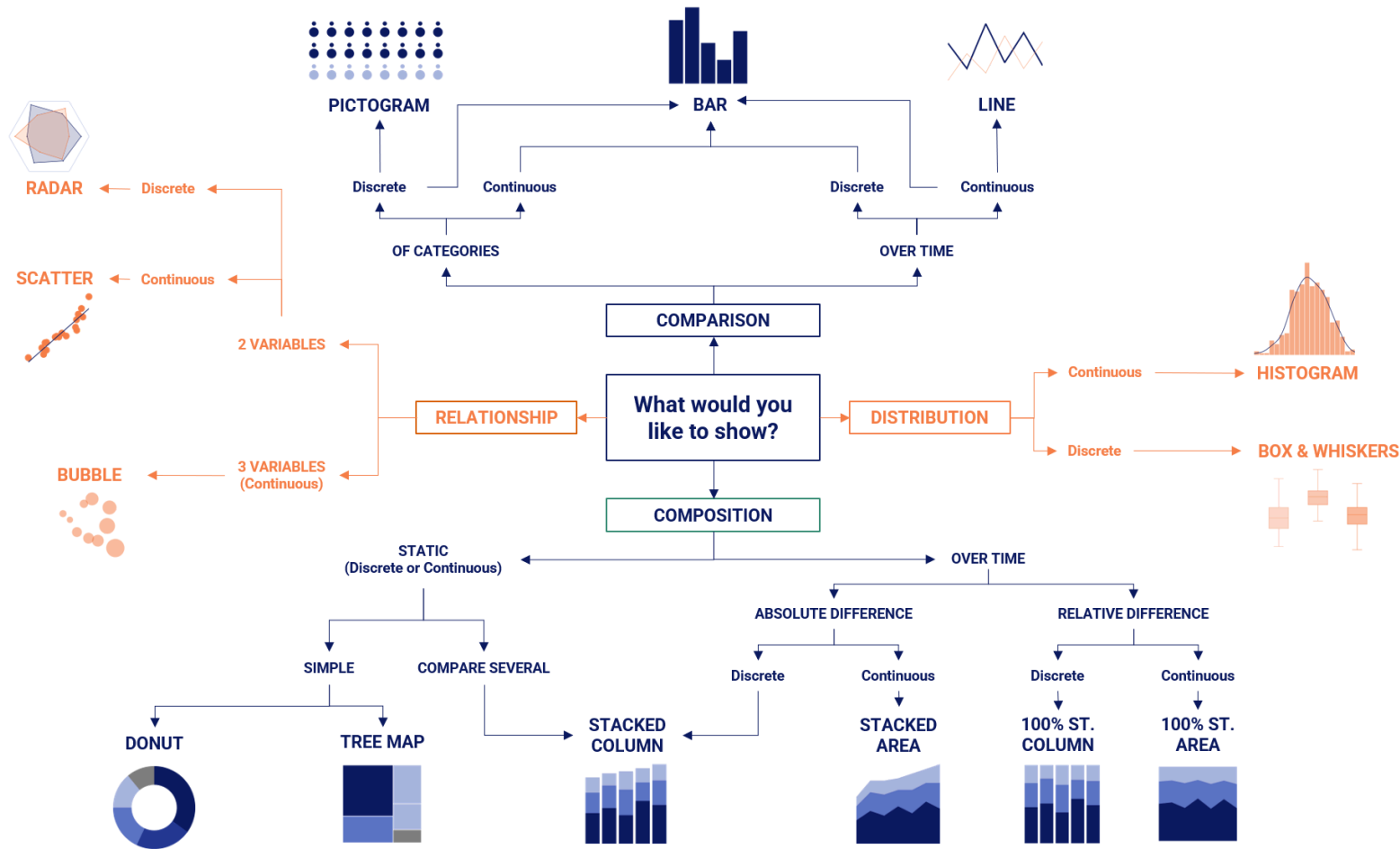
FT Visual Vocabulary

Organised by analytical purpose: nine families, 59 chart types, drawn from real editorial work.

Both ask the same question first: **what does your data need to say?**

These are guides, not rules written in stone. and there are many alternative frameworks.

2. WHAT DOES YOUR DATA NEED TO SAY?



THE ABELA'S DATA-TYPE DECISION TREE

One question at the centre — **what would you like to show?** — and four branches out of it.

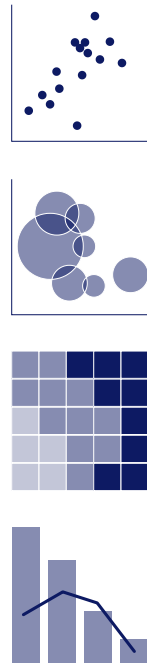
2. WHAT DOES YOUR DATA NEED TO SAY?

FINANCIAL TIMES VISUAL VOCABULARY

DEVIATION



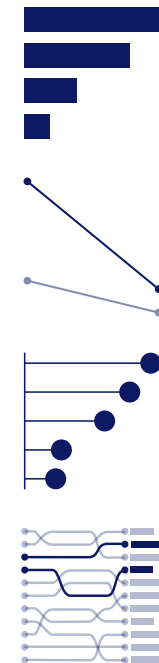
CORRELATION



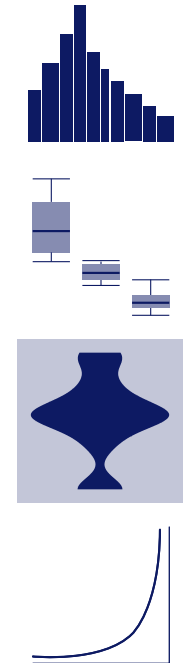
CHANGE V TIME



RANKING



DISTRIBUTION



Financial Times. (n.d.). *Visual vocabulary*. <https://ft-interactive.github.io/visual-vocabulary/>

2. WHAT DOES YOUR DATA NEED TO SAY?

FINANCIAL TIMES VISUAL VOCABULARY (CONT.)

PART OF WHOLE



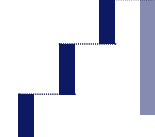
MAGNITUDE



SPATIAL



FLOW



Financial Times. (n.d.). *Visual vocabulary*. <https://ft-interactive.github.io/visual-vocabulary/>

3. WHO IS THE AUDIENCE AND WHAT DO THEY NEED?

THE ANALYST

Likes to see details and understand exactly what's happening at a granular level.

- Tables or combo charts
- Granular detail to support root-cause analysis

THE MANAGER

Wants summarized data with clear, actionable insights to help operate the business.

- Common charts & graphs
- Some detail, but only when it supports a specific insight

THE EXECUTIVE

Needs high-level, crystal clear KPIs to track business health and topline performance.

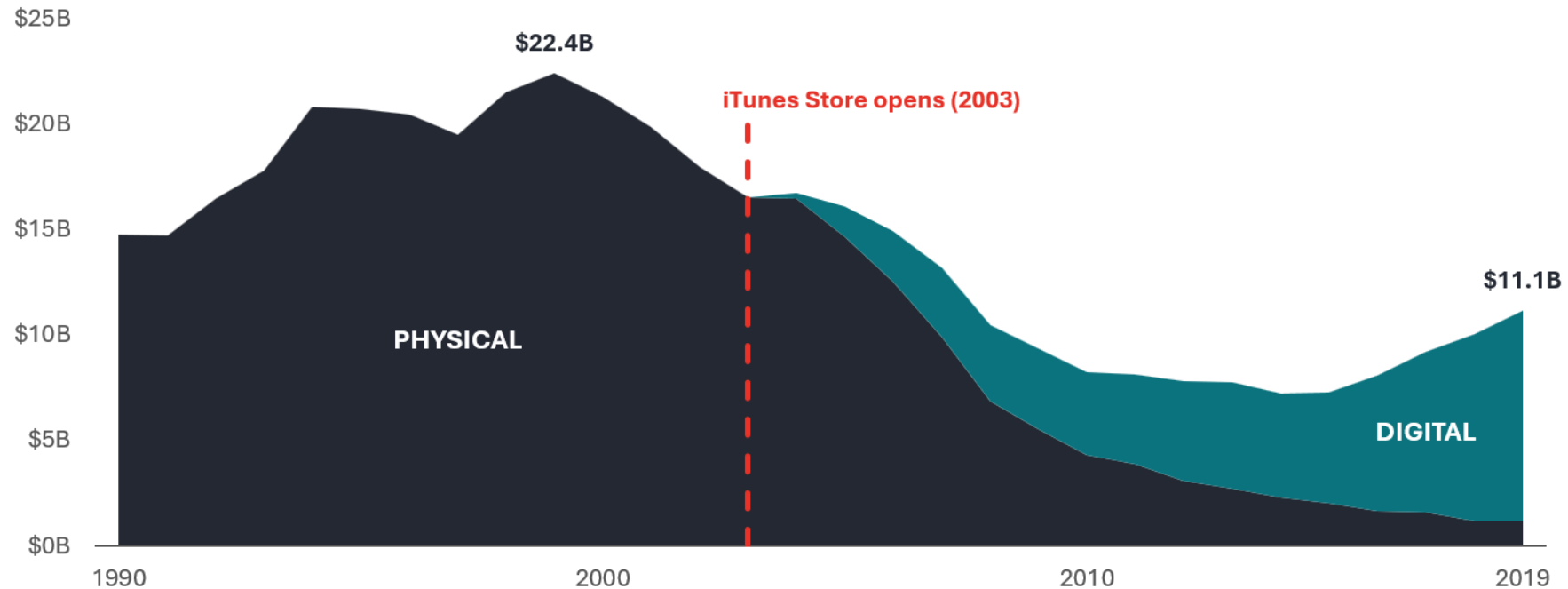
- KPI cards or simple charts
- Minimal detail, unless it adds critical context to KPIs

3. WHO IS THE AUDIENCE AND WHAT DO THEY NEED?

SAME DATA, SAME TAKEAWAY — TWO READERS

Digital revenue climbed steadily with more internet adoption

However, the music industry **shrunk 50%** comparing to 1999



EXECUTIVE — ONE MESSAGE, TWO NUMBERS

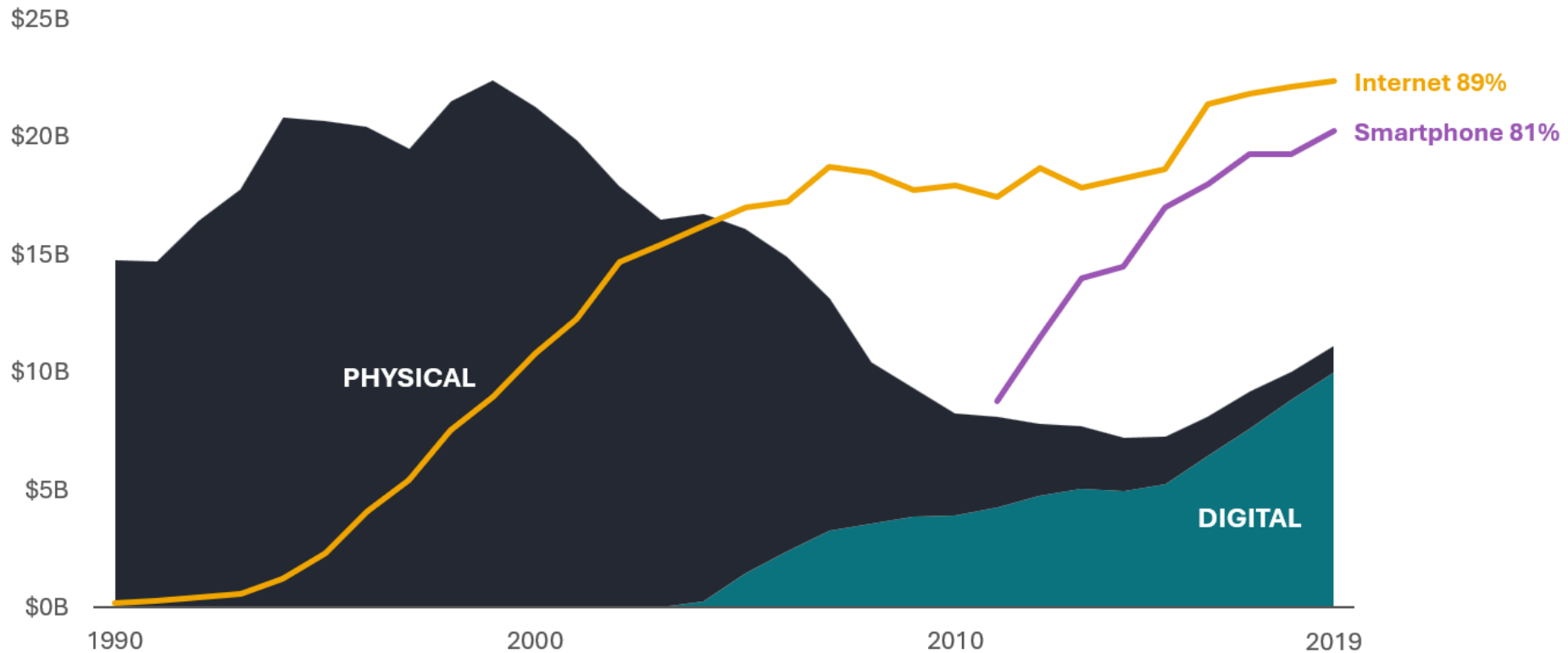
Source: RIAA U.S. Music Revenue Database, adjusted to 2019 dollars · adoption: World Bank / ITU and Pew Research Center.

3. WHO IS THE AUDIENCE AND WHAT DO THEY NEED?

SAME DATA, SAME TAKEAWAY — TWO READERS

Digital revenue climbed steadily with more internet adoption

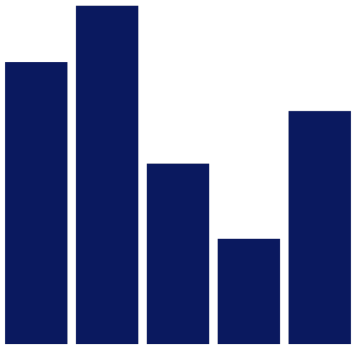
However, the music industry **shrunk 50%** comparing to 1999



ANALYST — THE DRIVER ALONGSIDE THE OUTCOME

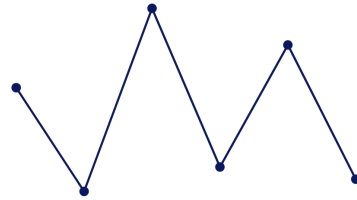
ALL THE CHARTS YOU'LL EVER NEED - 80% OF THE WAY

BAR / COLUMN CHART



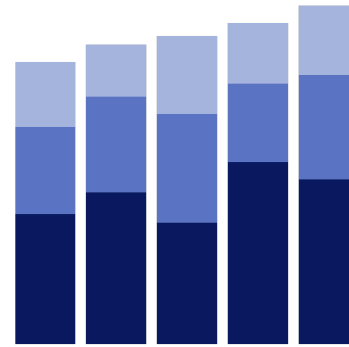
Used to show **comparisons** between categories

LINE CHART



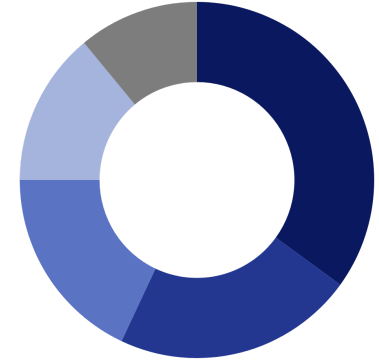
Used to show **trends** over time

STACKED BAR / COLUMN



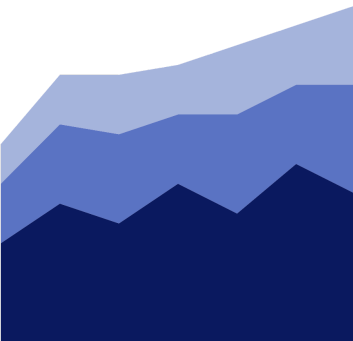
Used to show **composition** across categories

PIE / DONUT CHART



Used to show composition as **parts of a whole**

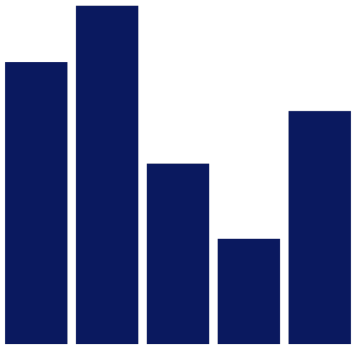
AREA CHART



Used to show **changes in composition** over time

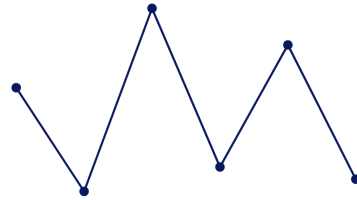
ALL THE CHARTS YOU'LL EVER NEED - 95% OF THE WAY

BAR / COLUMN CHART



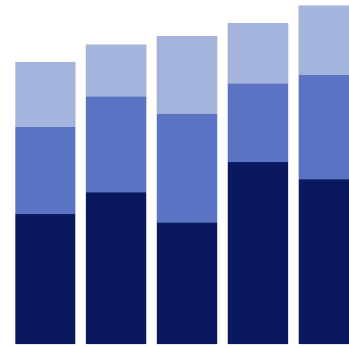
Used to show **comparisons** between categories

LINE CHART



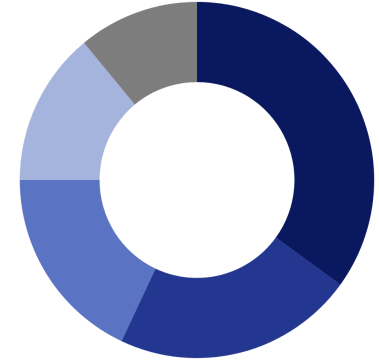
Used to show **trends** over time

STACKED BAR / COLUMN



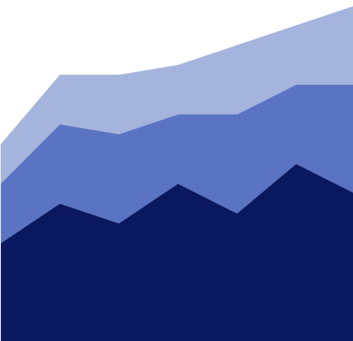
Used to show **composition** across categories

PIE / DONUT CHART



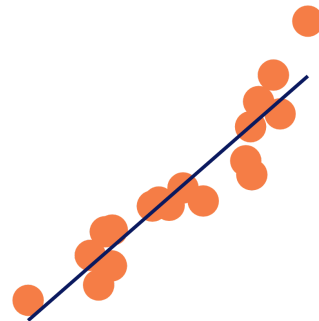
Used to show composition as **parts of a whole**

AREA CHART



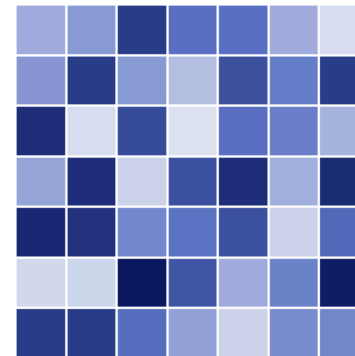
Used to show **changes in composition** over time

SCATTER PLOT



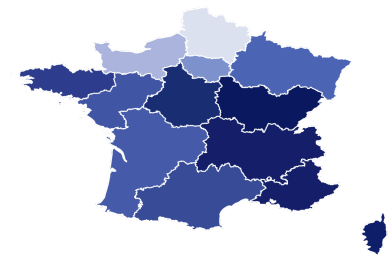
Used to show **relationships** between variables

TABLE / MATRIX



Used to show **numerical values** or highlight patterns

FILLED MAP



Used to visualise **geospatial** data

TIPS FOR EFFECTIVE VISUALIZATION

1 Choose the right *[basic]* chart

Functionality? Familiarity?

2 Substance over form

Pretty charts is NOT always good charts

3 Declutter, declutter, decluter

Delete until the next delete causes lacking of context

4 Have a message

Communicate an insight! *[but max 2 insights]*

5 Context matters

Enough to understand? Answer to the question?

6 Colors

50 Shades of Grey + ONE [max THREE] colors + shades

TIPS FOR EFFECTIVE VISUALIZATION

CHOOSE THE RIGHT *[basic]* CHART

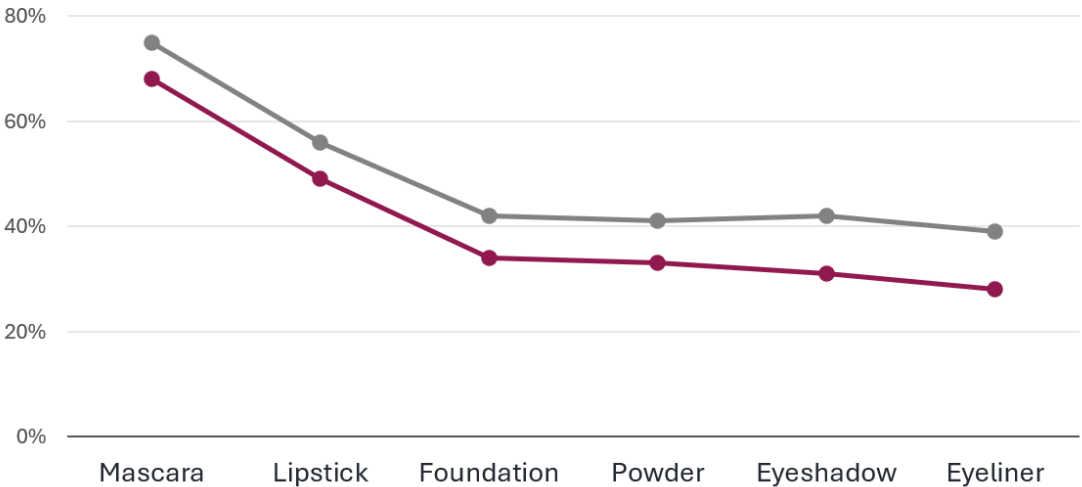
THE DATA

Type	2024	2019
Mascara	68%	75%
Lipstick	49%	56%
Foundation	34%	42%
Powder	33%	41%
Eyeshadow	31%	42%
Eyeliner	28%	39%

LINE — THE WRONG CHART

Daily makeup use: 2024 vs. 2019

% REPORTING DAILY USE



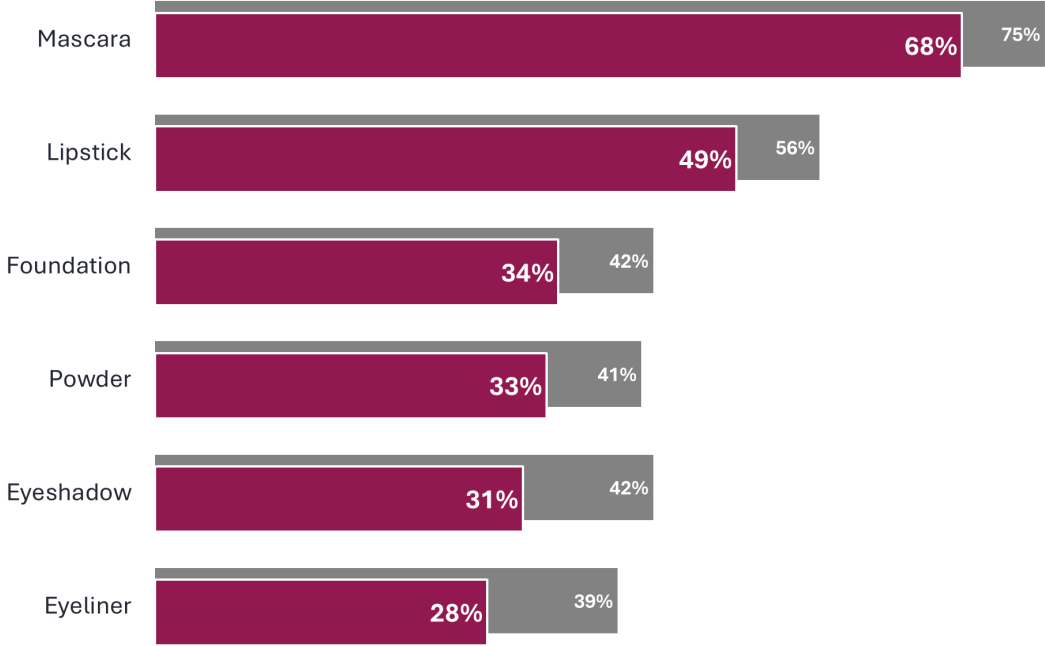
Lines usually indicate trends over time, which is not the main focus of this data

TIPS FOR EFFECTIVE VISUALIZATION

CHOOSE THE RIGHT *[basic]* CHART

Daily makeup use: 2024 vs. 2019

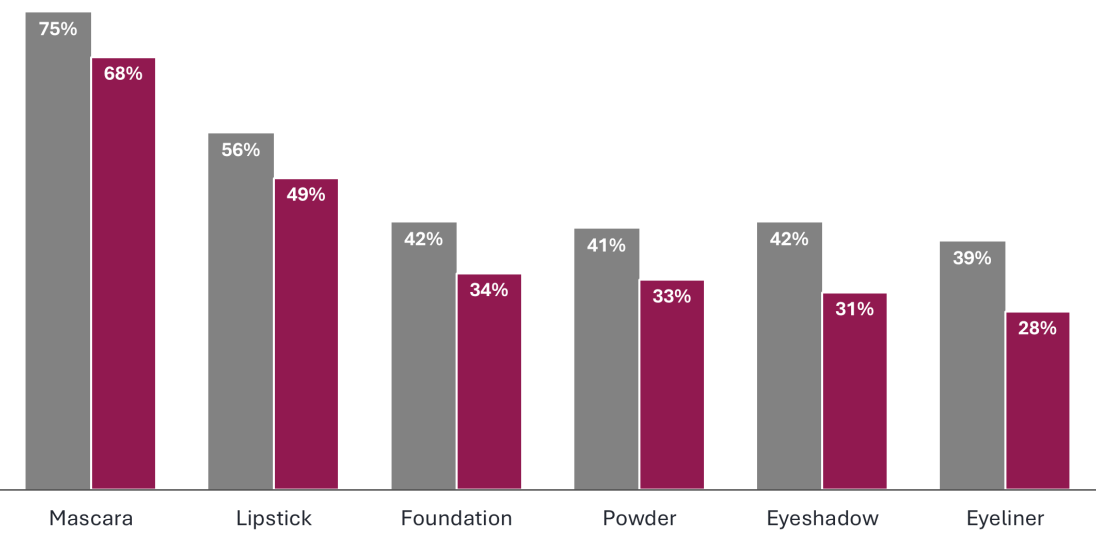
% REPORTING DAILY USE



OVERLAPPED BARS

Daily makeup use: 2019 vs. 2024

% REPORTING DAILY USE



COLUMNS, SIDE BY SIDE

Bar charts are almost always a safe choice

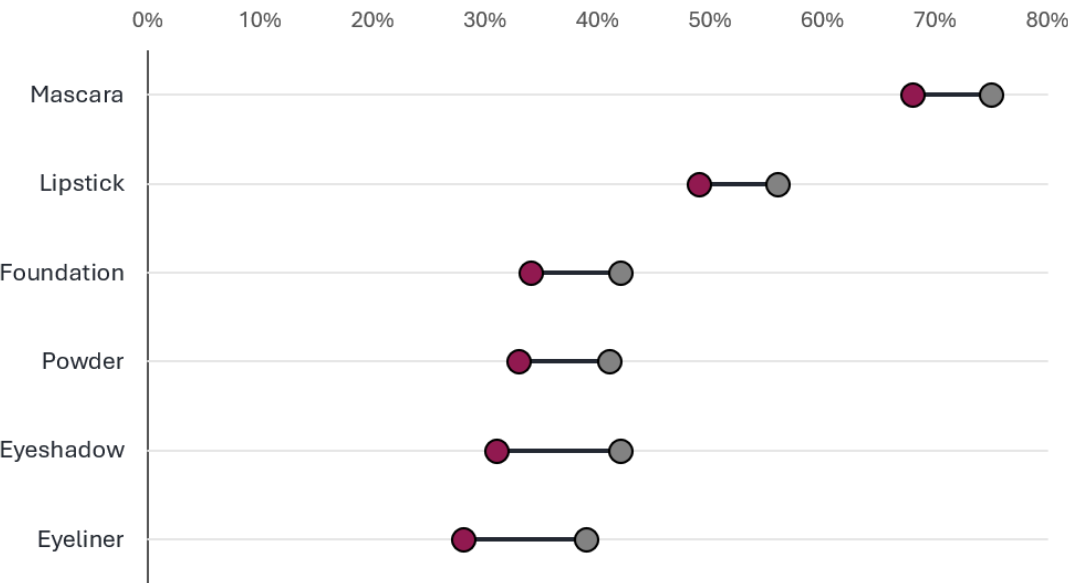
TIPS FOR EFFECTIVE VISUALIZATION

CHOOSE THE RIGHT *[basic]* CHART

DOT PLOT — THE GAP

Daily makeup use: **2024** vs. **2019**

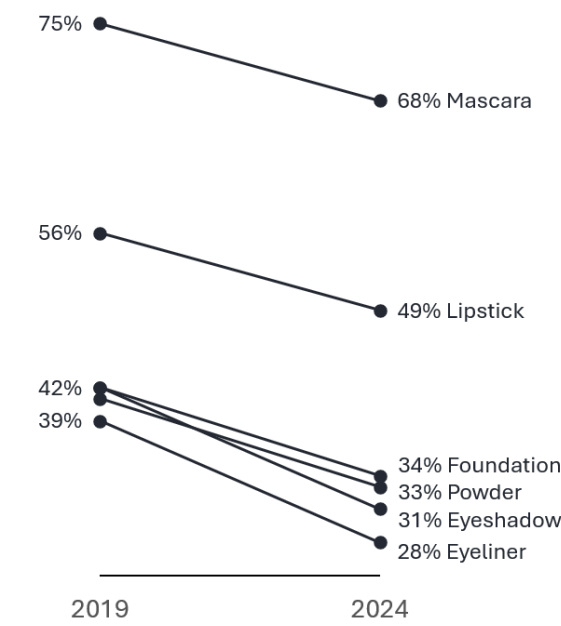
% REPORTING DAILY USE



SLOPEGRAPH — THE MOVE

Daily makeup use

% REPORTING DAILY USE



It looks nice, but familiarity might be an issue. Readers spend more effort than the well-known bars.

TIPS FOR EFFECTIVE VISUALIZATION

SUBSTANCE OVER FORM

Boston holds half the company's headcount



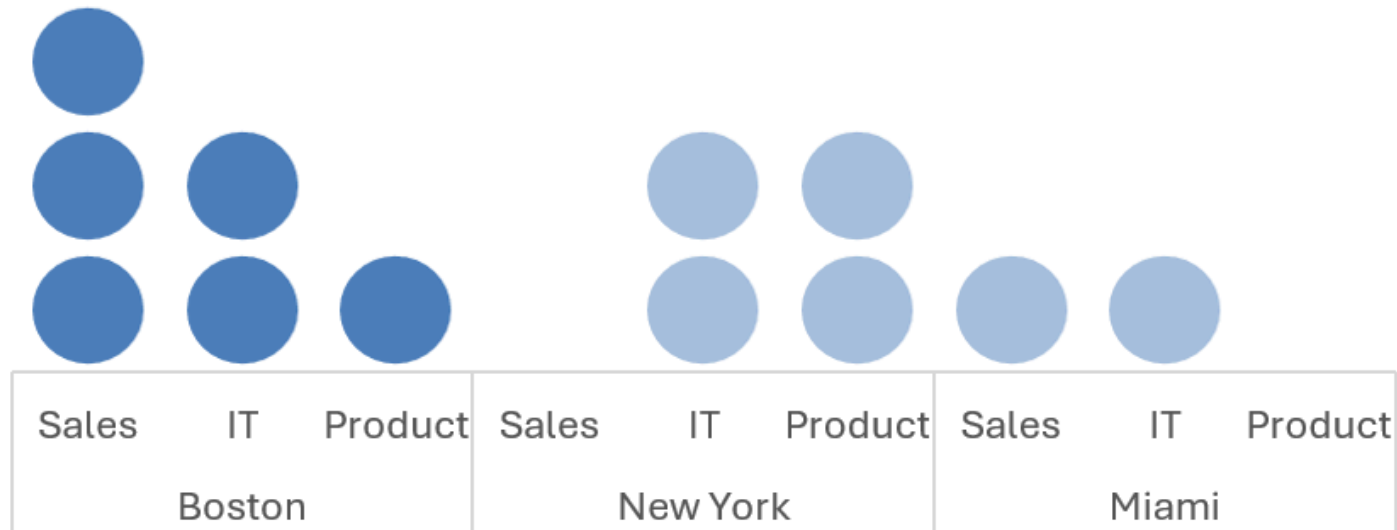
PRETTY

- But **every name is bent around its arc** — twelve of them, and you tilt your head for each one.
- Size is carried by **angle**.
- And it **cannot draw a zero**. Miami has no Product team.

TIPS FOR EFFECTIVE VISUALIZATION

SUBSTANCE OVER FORM

Boston holds half the company's headcount



READABLE

- One circle, one person. You count rather than estimate.
- Teams sit on a **shared baseline**, so Boston Sales against Miami Sales is a straight comparison down one line.
- The cost is honest: it answers *how many*, not *who*.

TIPS FOR EFFECTIVE VISUALIZATION

SUBSTANCE OVER FORM

Office	Team	Employee
Boston	Sales	Lisa T.
Boston	Sales	Phil B.
Boston	Sales	Marc G.
Boston	IT	And.
Boston	IT	Lauren W.
Boston	Product	Bill B.
New York	IT	Wendy L.
New York	IT	Gina C.
New York	Product	John R.
New York	Product	Darren S.
Miami	Sales	Vicky A.
Miami	IT	Dennis M.

OR JUST THE TABLE

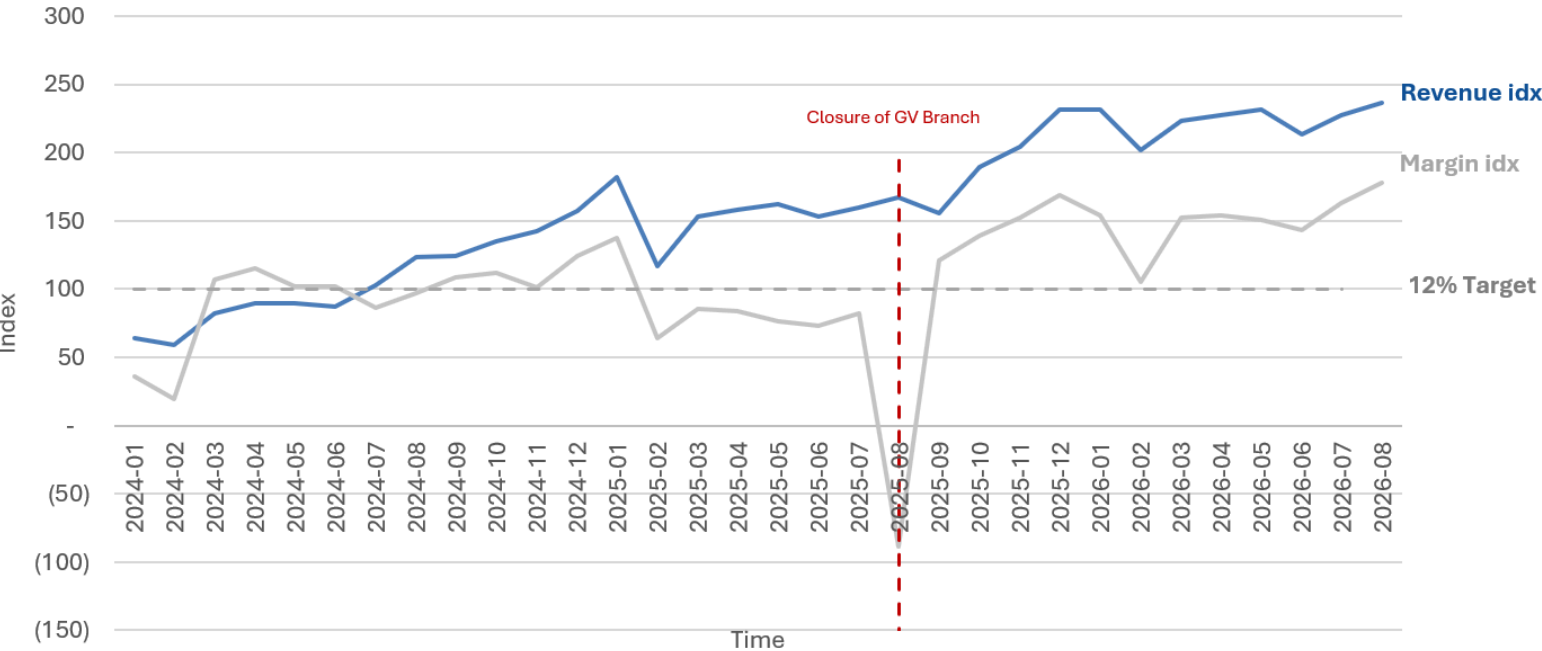
- A chart earns its place by making a comparison easier than reading the numbers would be. **At this size, a table does just fine.**

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

Revenue and Margin moved in line except for closure of Gò Vấp

Revenue vs Margin, 2024 avg revenue - 100, 12% margin - 100



BEFORE

Everything is competing for space.

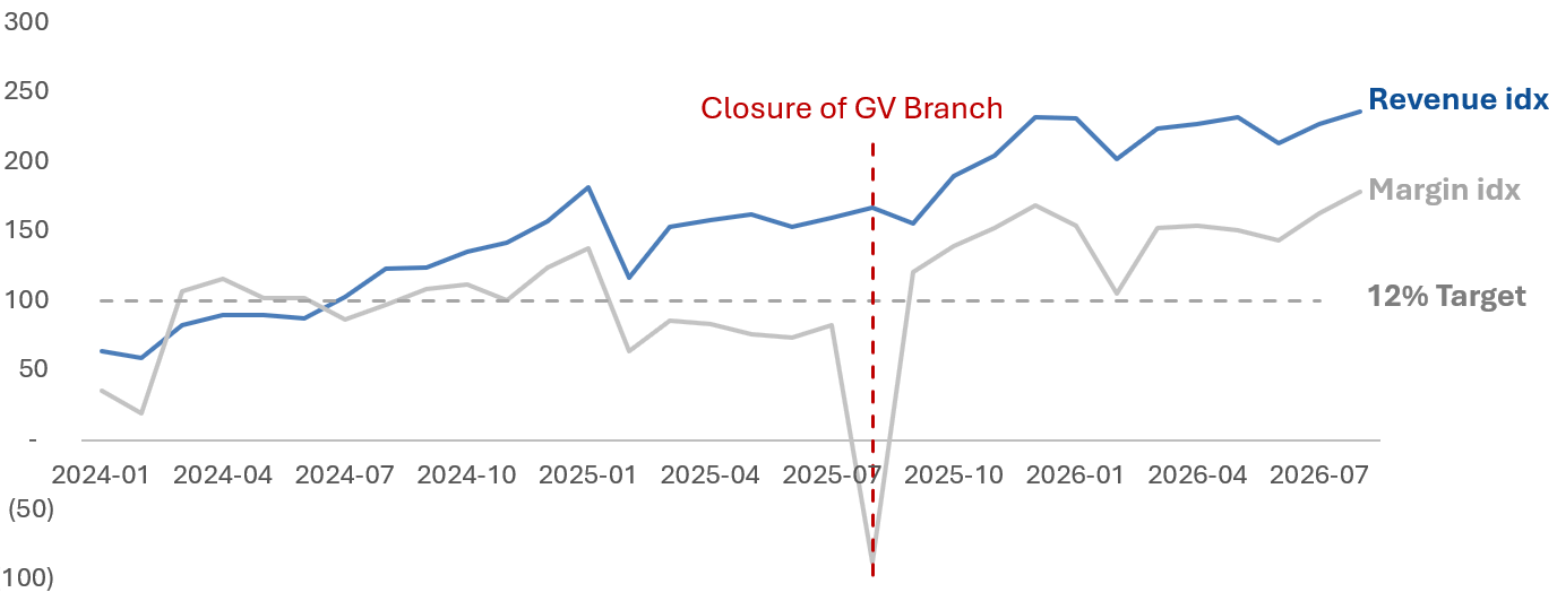
- **Gridlines** sit behind the data.
- **Two axis titles** repeat what the tick labels already say.
- **Thirty-two month labels**, rotated vertical because they will not fit any other way.

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

Revenue and Margin moved in line except for closure of Gò Vấp

Revenue vs Margin, 2024 average revenue - 100, 12% profit margin - 100



ONE PASS

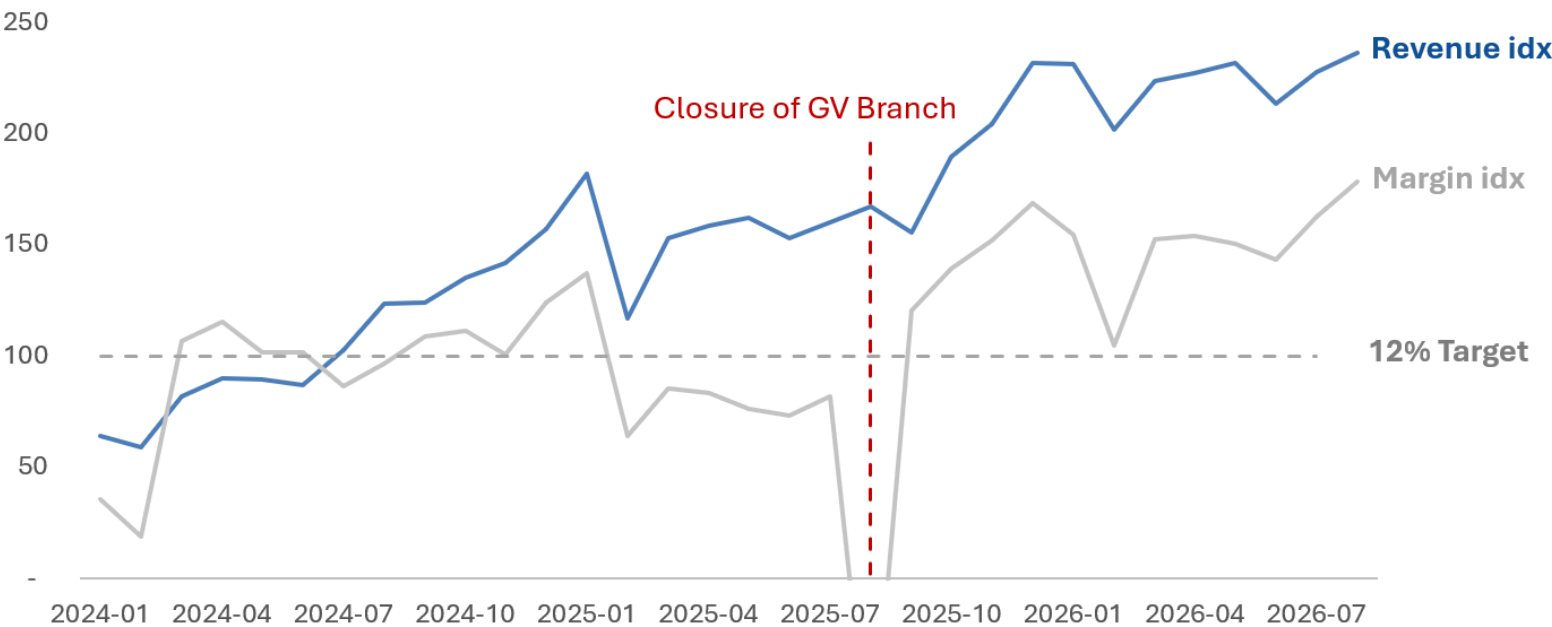
Gridlines, axis titles and tick marks gone; months thinned to every third and set horizontal.

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

Revenue and Margin moved in line except for closure of Gò Vấp

Revenue vs Margin, 2024 average revenue - 100, 12% profit margin - 100



TOO FAR?

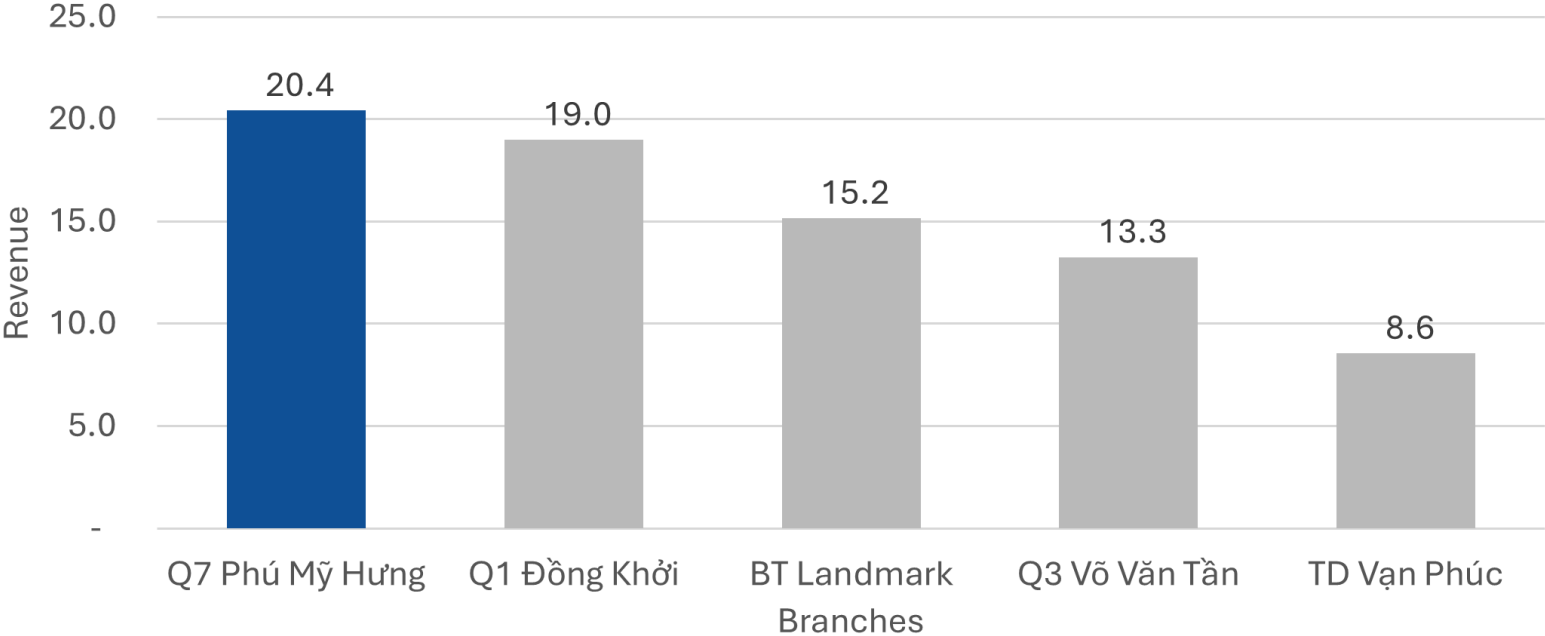
The axis is tightened to 0-250, which lifts the data up into the frame. **Does the exact measurement of the loss month important?**

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

**Phú Mỹ Hưng generated the most revenues follow by
Đồng Khởi**

Revenues by branch 2026 YTD - VNDbil



THE DEFAULT

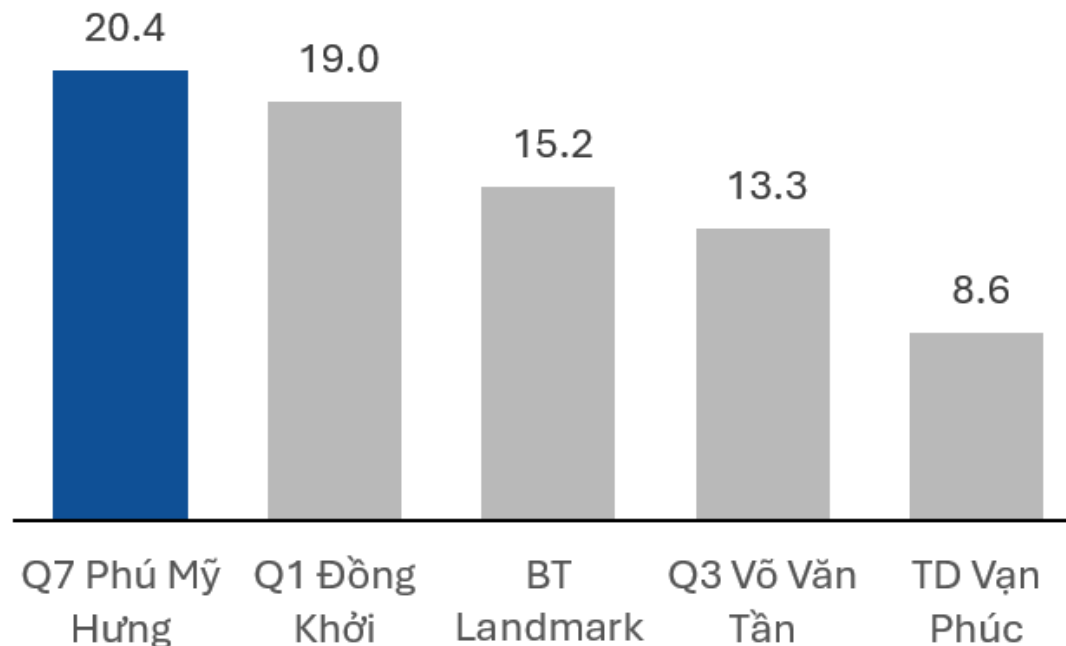
- **Gridlines** behind every bar.
- **Two axis titles** — *Revenue* and *Branches* — naming what the labels beside them already name.
- **Every number twice:** once up the axis, once on the bar.

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

Phú Mỹ Hưng generated the most revenues follow by Đồng Khởi

Revenues by branch 2026 YTD - VNDbil



ONE PASS

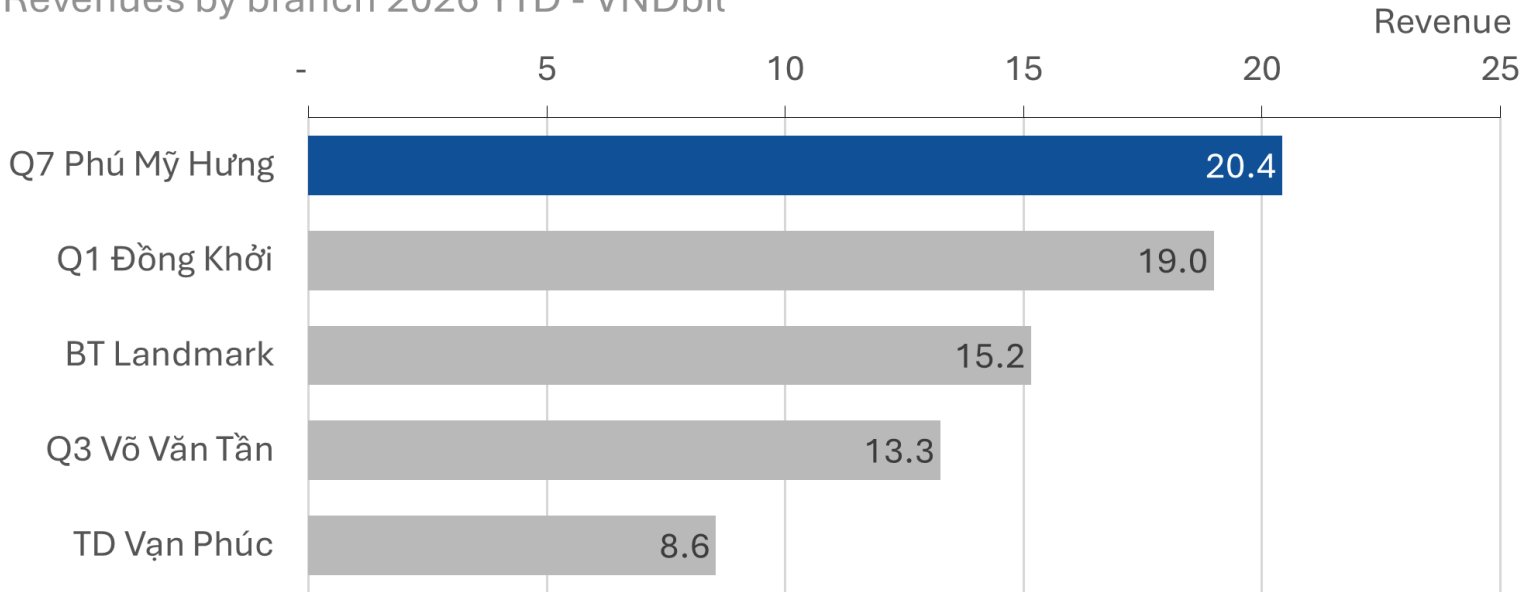
- Gridlines, value axis and both titles gone. The labels on the bars keep the numbers.
- One bar in colour, the rest grey — colour now means something.

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

**Phú Mỹ Hưng generated the most revenues follow by
Đồng Khởi**

Revenues by branch 2026 YTD - VNDbil



MAYBE HORIZONTAL?

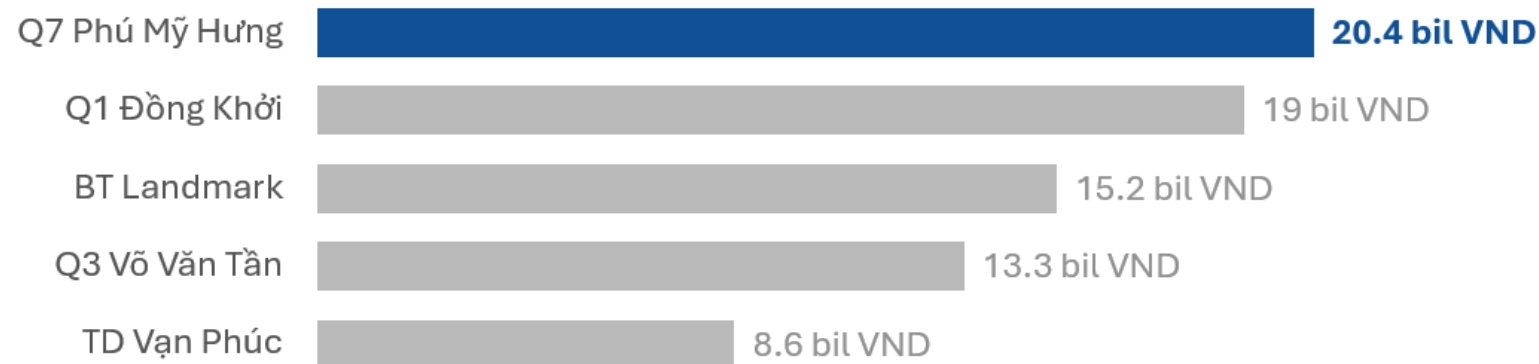
- Horizontal bars give each name a full line of its own. Nothing wraps.

TIPS FOR EFFECTIVE VISUALIZATION

DECLUTTER, DECLUTTER, DECLUTTER

Phú Mỹ Hưng generated the most revenues follow by Đồng Khởi

Revenues by branch 2026 YTD



DECLUTTERING

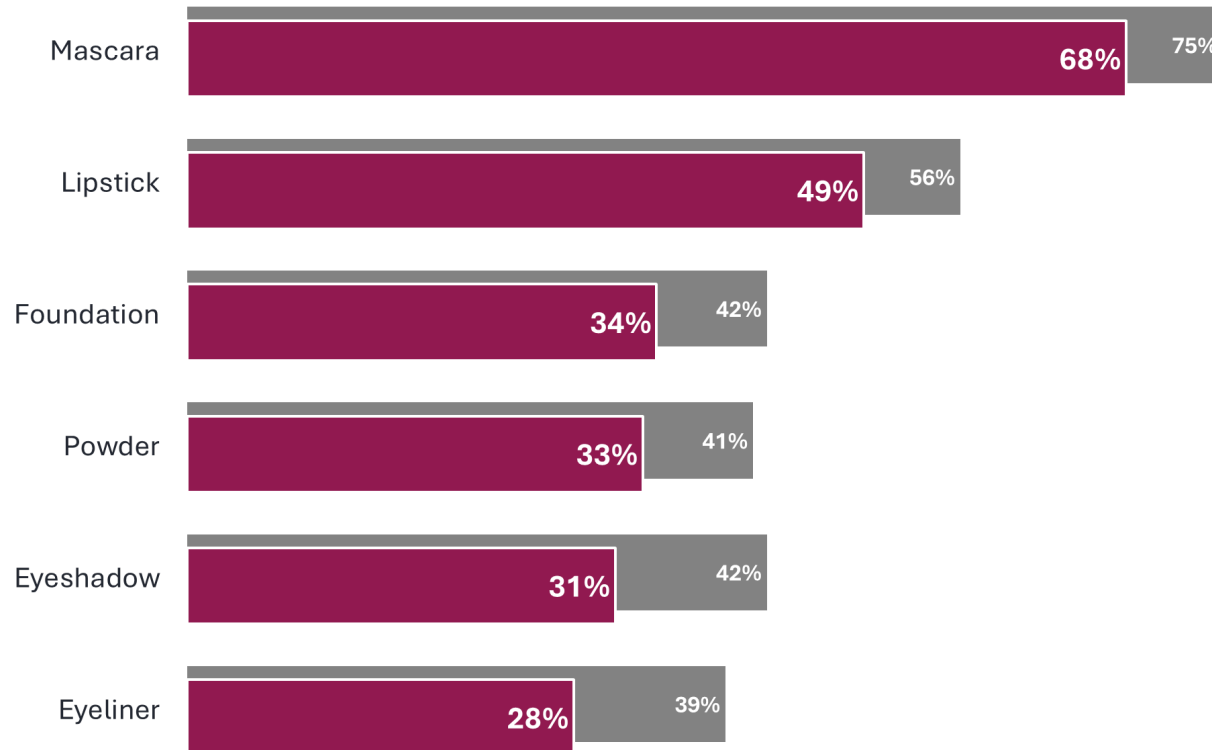
- Axis, ticks and gridlines gone. Each bar is labelled where it ends.
- **Units stated once**, on the bar that carries the message.
- Sorted, one bar blue. The picture now **makes the title's point** instead of leaving the reader to go and find it.

TIPS FOR EFFECTIVE VISUALIZATION

HAVE A MESSAGE

Daily makeup use: **2024** vs. 2019

% REPORTING DAILY USE



DO NOT let the reader scramble for insight on their own.

WHAT'S THE MESSAGE?

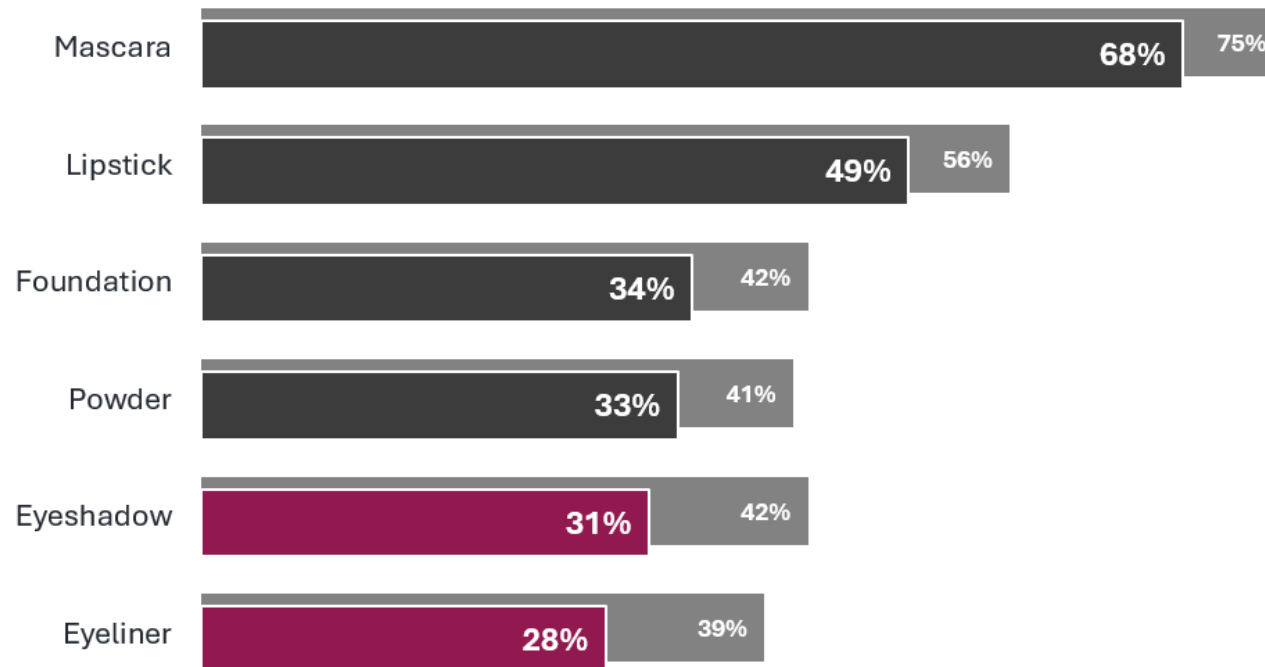
- **All six fell.** Not one category held its 2019 level.
- **The drops pair off by effort:** mascara and lipstick –7 points, foundation and powder –8, eyeshadow and eyeliner –11.
- **Mascara and lipstick still lead.**
- **Eyeshadow and eyeliner** recorded sharpest declines..

TIPS FOR EFFECTIVE VISUALIZATION

HAVE A MESSAGE

Eyeshadow & Eyeliner recorded the sharpest declines

Daily makeup use: **2024** vs. **2019**. % reporting daily use



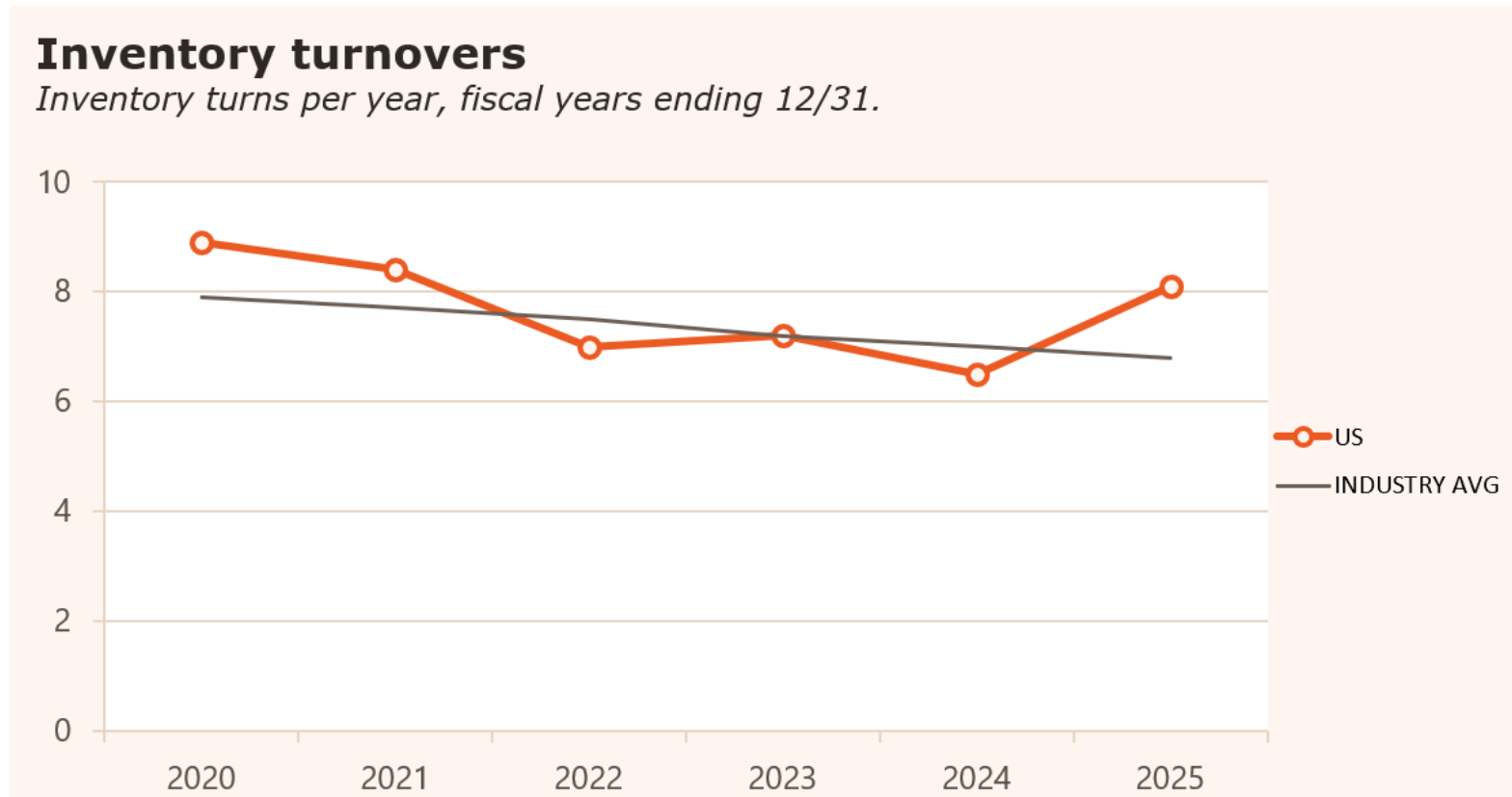
CLEAR MESSAGE

- Eyeshadow and eyeliner recorded sharpest declines..

The reader confirms the insight first. They might read further **on their own** but you **GURANTEED** you communicated your intended message.

TIPS FOR EFFECTIVE VISUALIZATION

HAVE A MESSAGE



WHAT'S THE MESSAGE?

- The recent years below industry average.
- Just recover to above industry average.

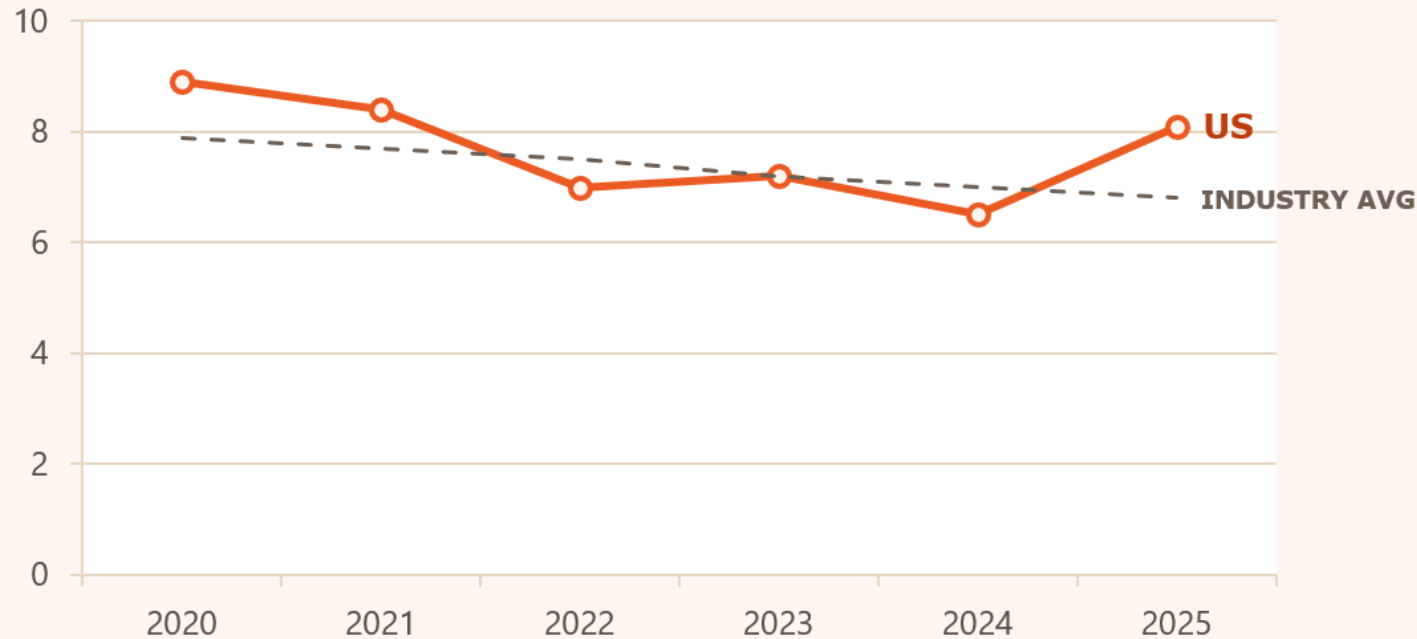
USE WORDING to clarify what you want to communicate.

TIPS FOR EFFECTIVE VISUALIZATION

HAVE A MESSAGE

We return to turning more inventory than the industry average after 3 years lacking behind

Inventory turns per year, fiscal years ending 12/31. Higher is better; the dashed line is the industry benchmark.



CLEAR MESSAGE

- The company turnover ratio returned to being above industry average after 3 years of lacking behind.

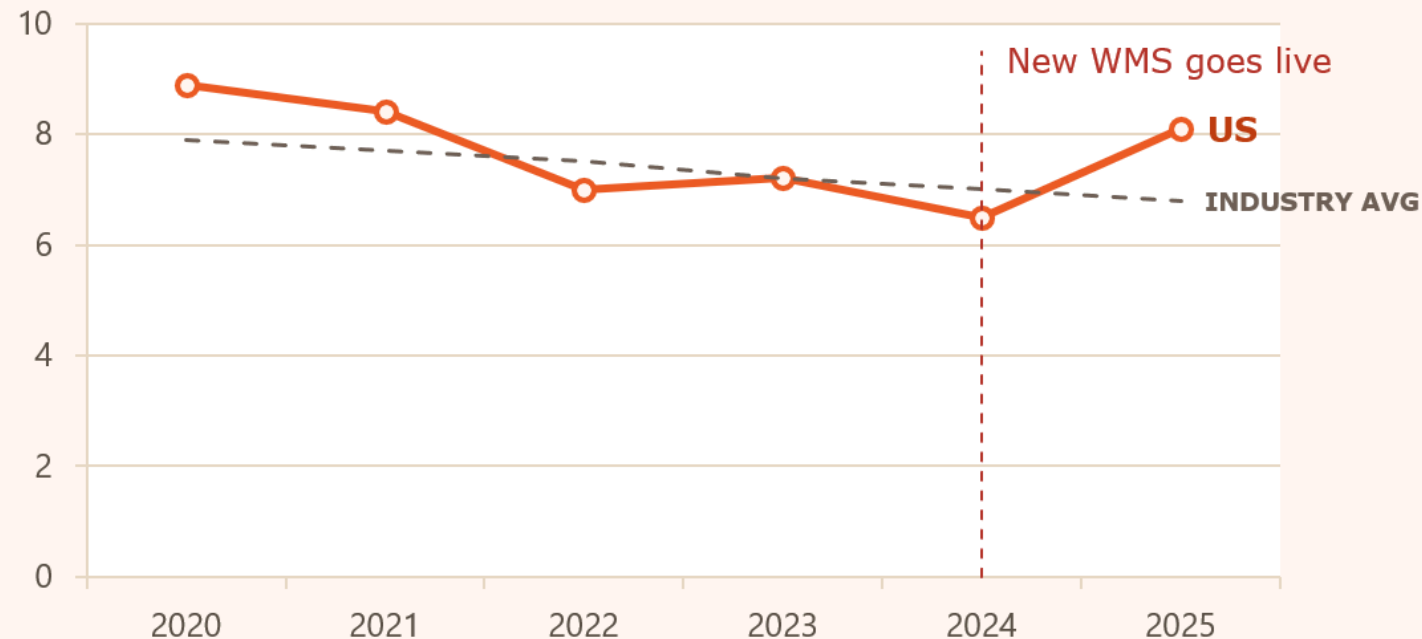
WORDING to ensure brief and correct message.

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

We return to turning more inventory than the industry average after 3 years lacking behind

Inventory turns per year, fiscal years ending 12/31. Higher is better; the dashed line is the industry benchmark.



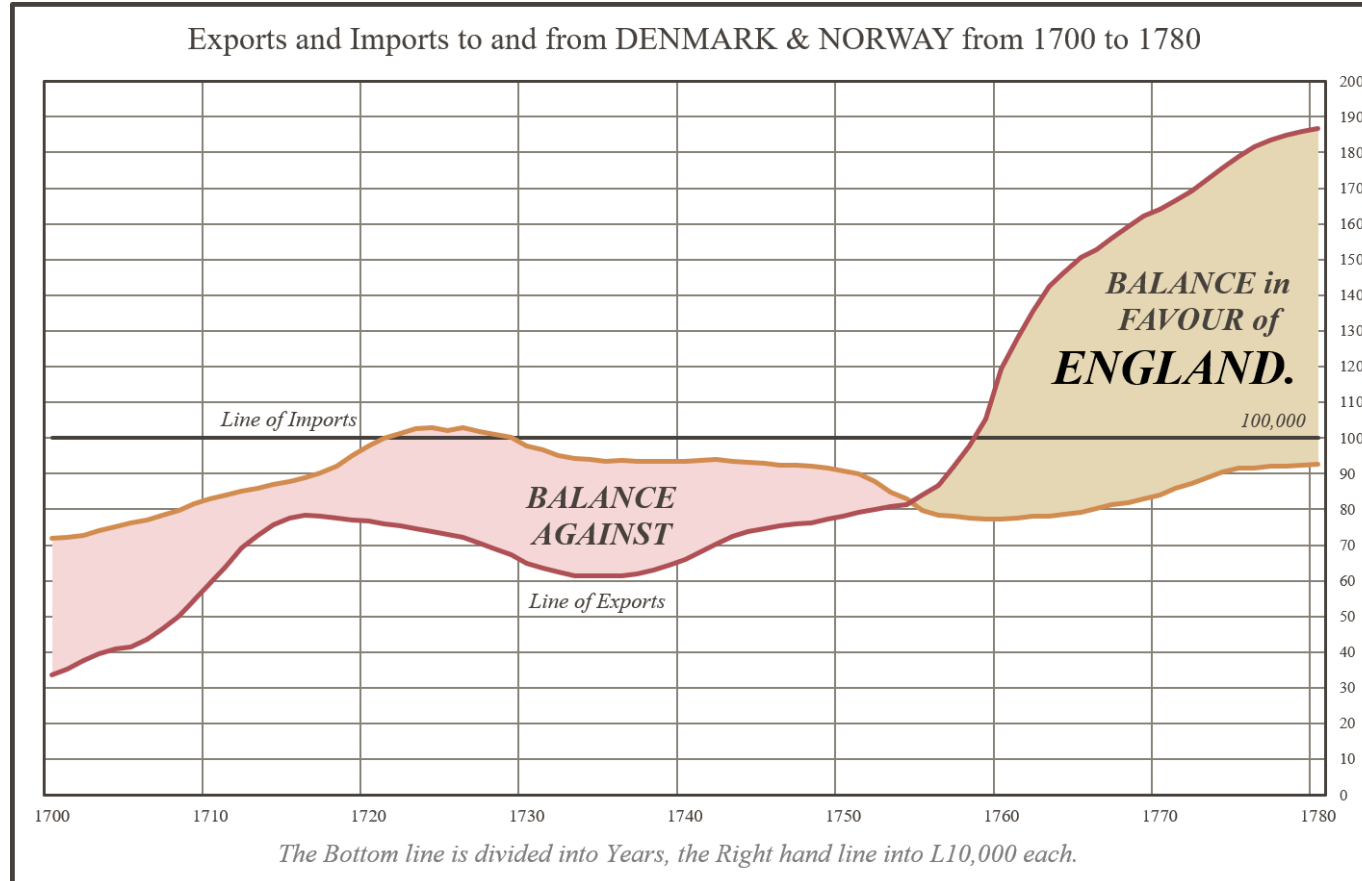
FACT WITH EXPLANATION

- The company turnover ratio returned to being above industry average after 3 years of lacking behind **thanks to** introduction of the new WMS system.

ADDITIONAL FEATURE to add to the context of the message.

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS



1ST EVER LINE CHART

- Blending context into the design **ENDLAND**
- **100,000** line - common quantity to accelerate recognition
- Use of **colors**

The chart was created in 1786. That's almost 350 years ago.

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

Feature	Have not used	Not satisfied at all	Not very satisfied	Somewhat satisfied	Very satisfied	Completely satisfied
Search	0%	1%	1%	11%	40%	47%
Mobile app	1%	2%	1%	13%	36%	47%
Notifications	2%	2%	5%	24%	34%	33%
File sharing	7%	2%	4%	21%	37%	29%
Calendar sync	5%	2%	6%	23%	36%	28%
Dashboards	13%	2%	5%	20%	35%	25%
Task comments	18%	3%	5%	15%	26%	33%
Templates	11%	3%	6%	23%	32%	25%
Time tracking	21%	3%	5%	17%	27%	27%
Bulk edit	2%	8%	14%	24%	27%	25%
Custom fields	27%	3%	4%	17%	28%	21%
Automations	27%	3%	4%	23%	27%	16%
Report export	33%	3%	8%	25%	18%	13%
Guest access	26%	9%	14%	24%	17%	10%
API access	50%	2%	6%	15%	16%	11%

THE DATA

- "How satisfied have you been with each of these features?" — 15 features, 6 answers, in % of respondents.
- Depending on the specific question/reader, different charts maybe utilised.

Everything that follows is **this same table** — only looked at differently.

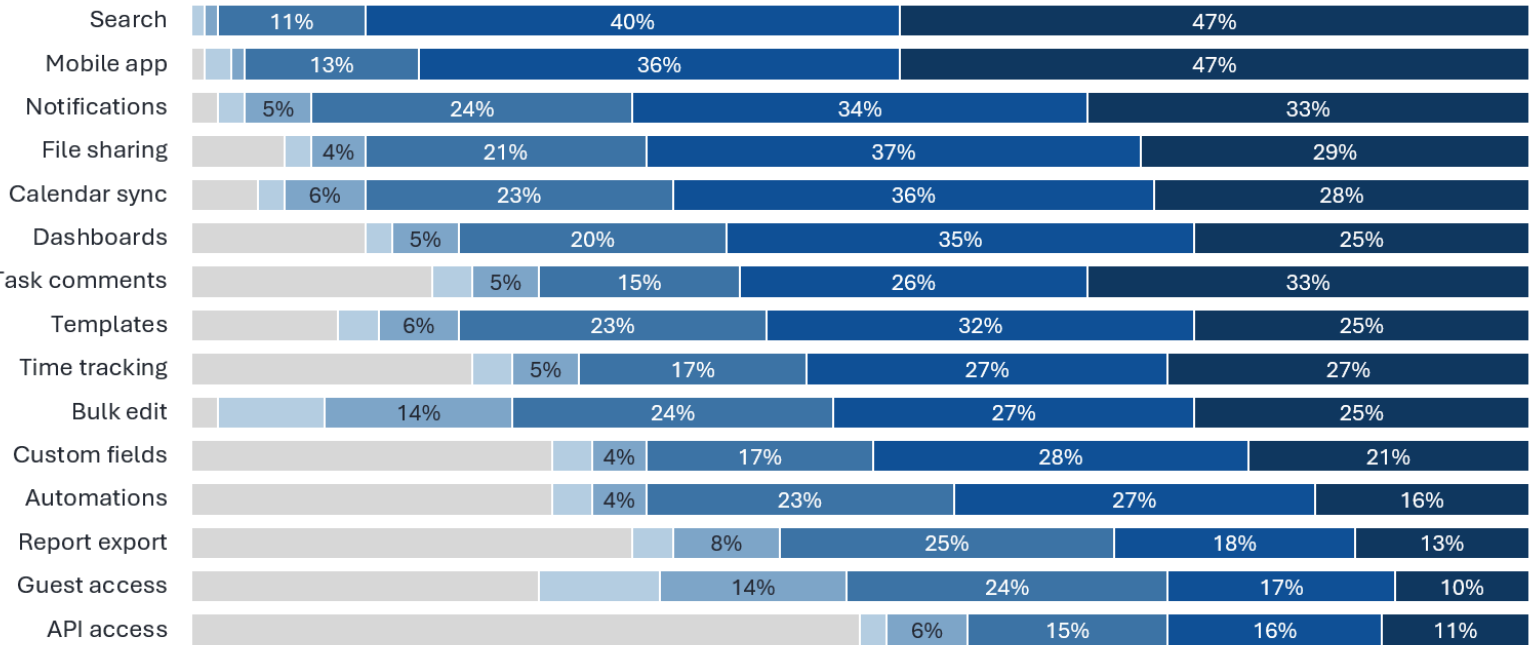
TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS - ONE CHART, MANY QUESTIONS

How satisfied have you been with each of these features?

Product X user satisfaction: features — % of respondents

■ Have not used ■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



WHAT'S THE MESSAGE?

- 15 features × 6 answers — 90 numbers, all shouting at once.
- Six shades of one colour: the reader has to keep going back to the legend.

A survey answers **several** questions. One picture should only make **one** point.

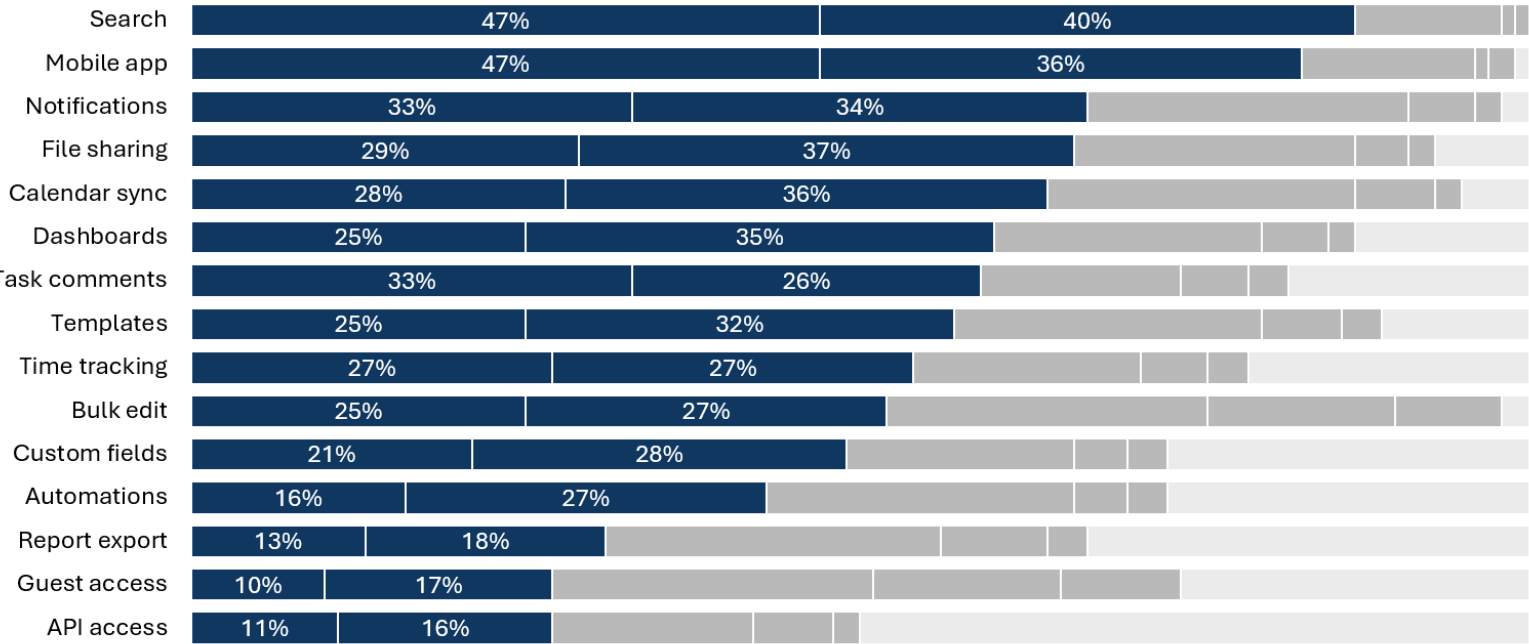
TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

SEARCH and MOBILE APP has highest user satisfaction

User satisfaction (%) by features

■ Completely satisfied ■ Very satisfied ■ Somewhat satisfied ■ Not very satisfied ■ Not satisfied at all ■ Have not used



CLEAR MESSAGE

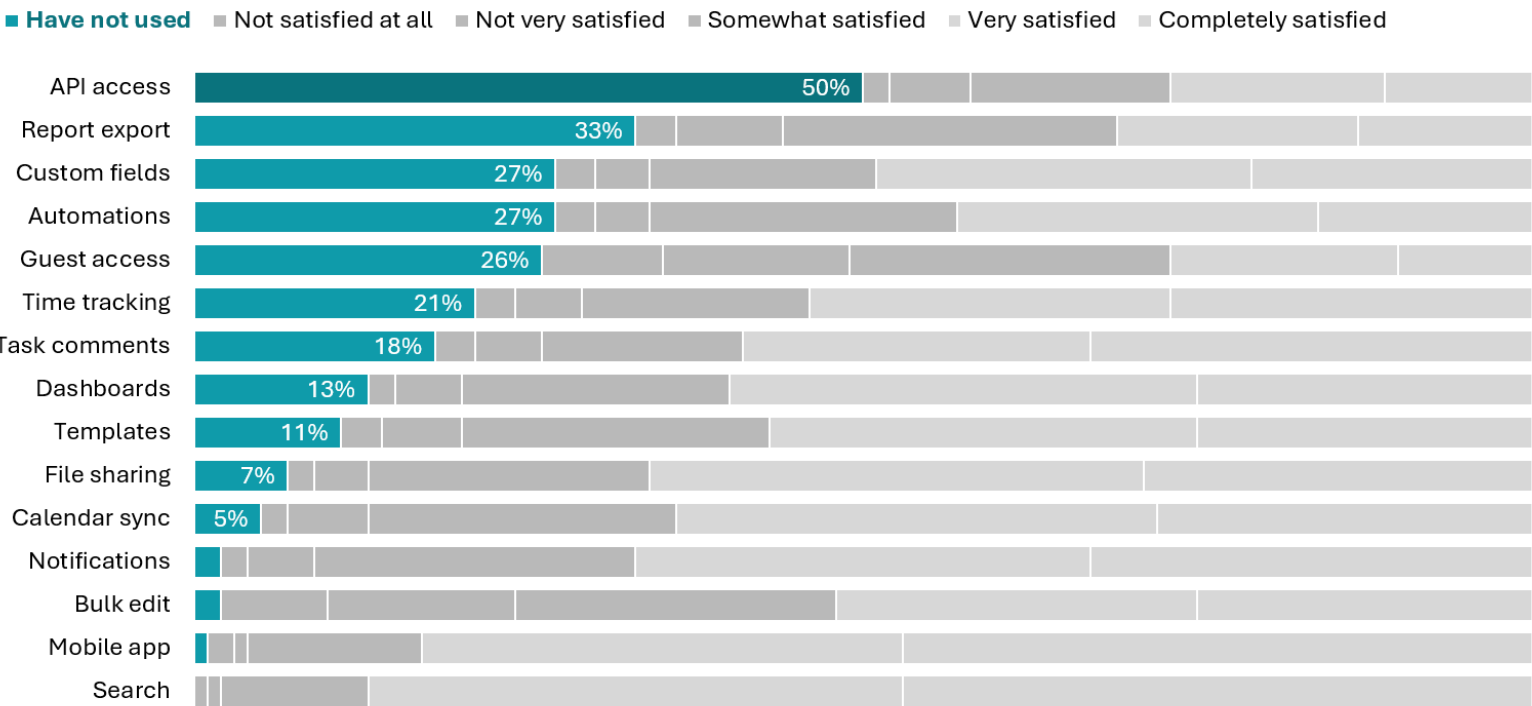
- Answer a question about **satisfaction**.
- The two satisfied answers are moved to the **left edge**, so every bar starts from the same line and the ranking is readable without arithmetic.
- The other four answers stay — in grey. **Nothing is hidden, only de-emphasised.**

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

API ACCESS is barely used — half have never touched it

Have not used the feature (%), by feature



NEW QUESTION

- Answer a question about **unutilised**.
- The order is **flipped**: what ranked last on unutilisation ranks first here.
- Move **focused metrics to the left**.

Move the answer you care about **to the axis**, and re-sort by it.

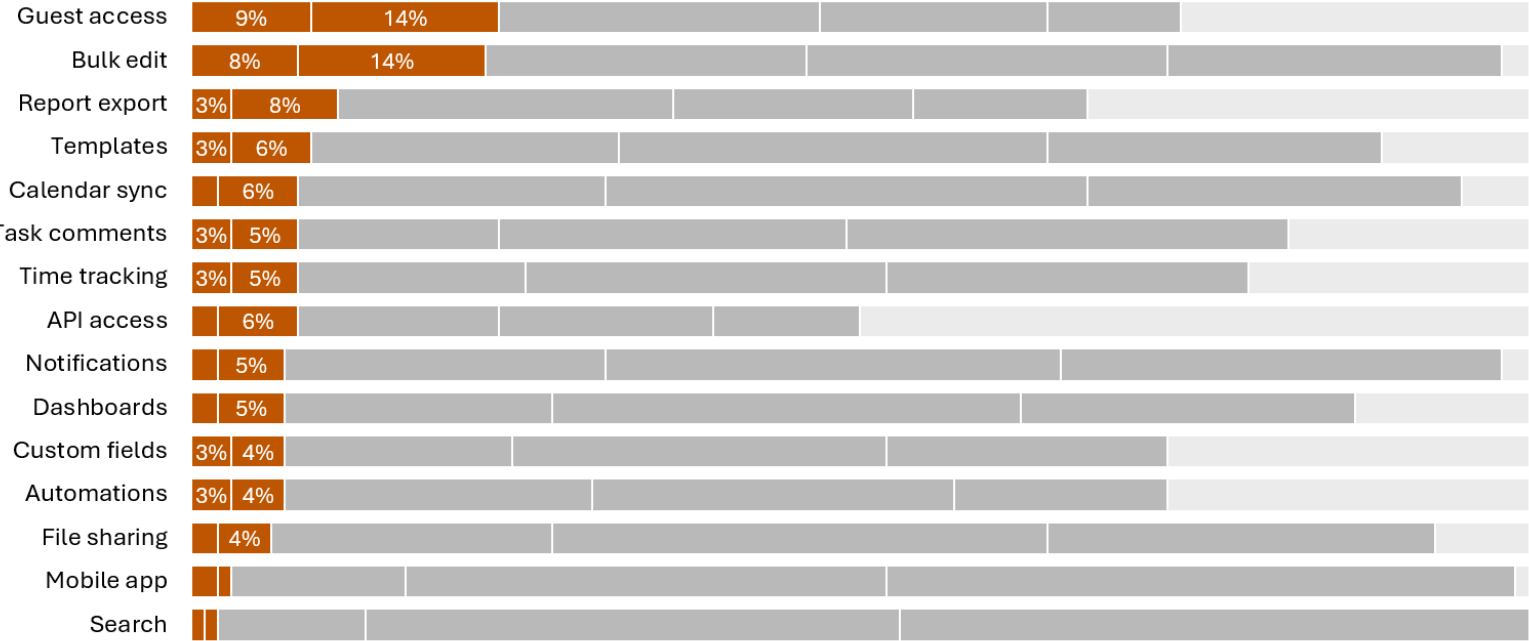
TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

GUEST ACCESS and BULK EDIT draw the most dissatisfaction

Not very or not at all satisfied (%) by feature

■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied ■ Have not used



CLEAR MESSAGE

- Answer a question about dissatisfaction.
- Noted we used different colors for different metrics.

TIPS FOR EFFECTIVE VISUALIZATION

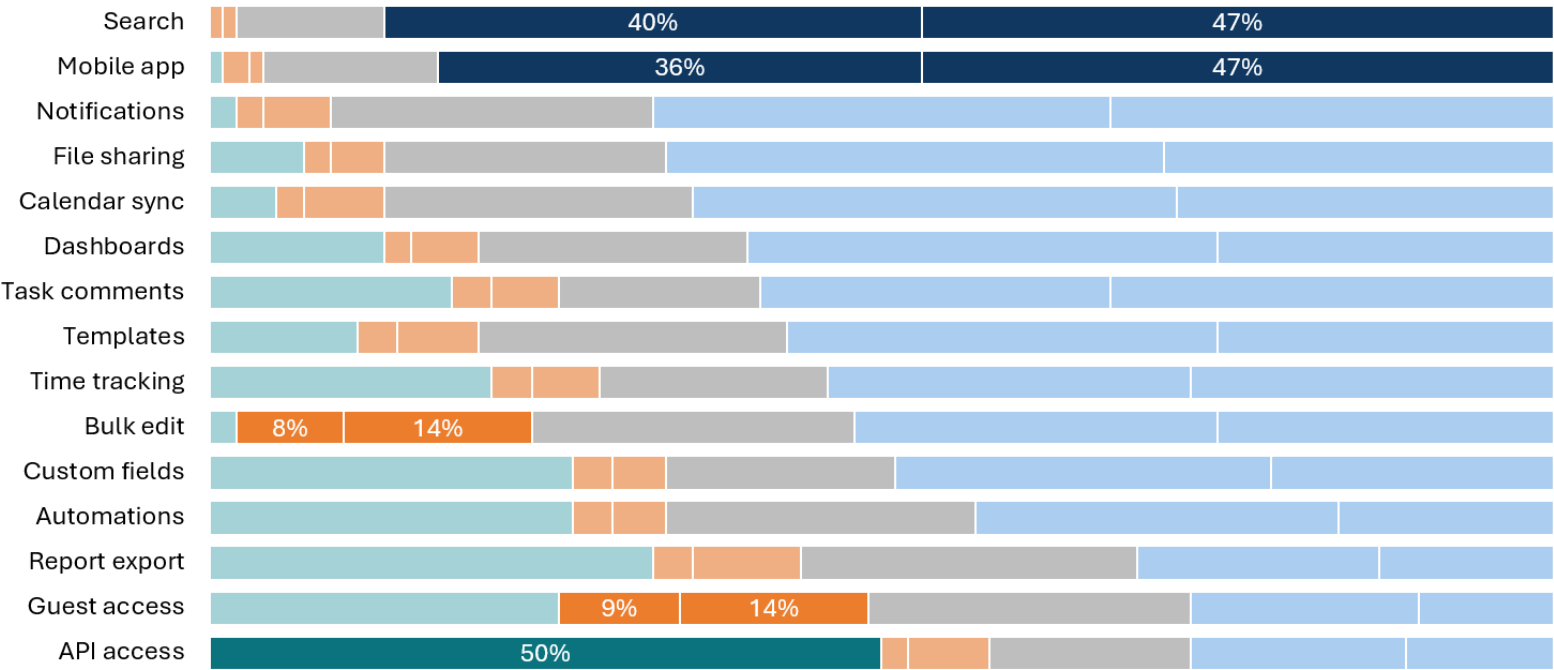
CONTEXT MATTERS

If you really have to include all three message in a

User satisfaction varies greatly by feature

Product X user satisfaction: features

■ Have not used ■ Not satisfied at all ■ Not very satisfied ■ Somewhat satisfied ■ Very satisfied ■ Completely satisfied



Search and the mobile app continue to top user satisfaction.

Users are least satisfied with bulk edit and guest access. What improvements can we make here for a better experience?

API access is least used — half have never touched it. What can we do to raise utilisation among existing users?

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

What if you need to know which feature to **fix first?** **promote more?**

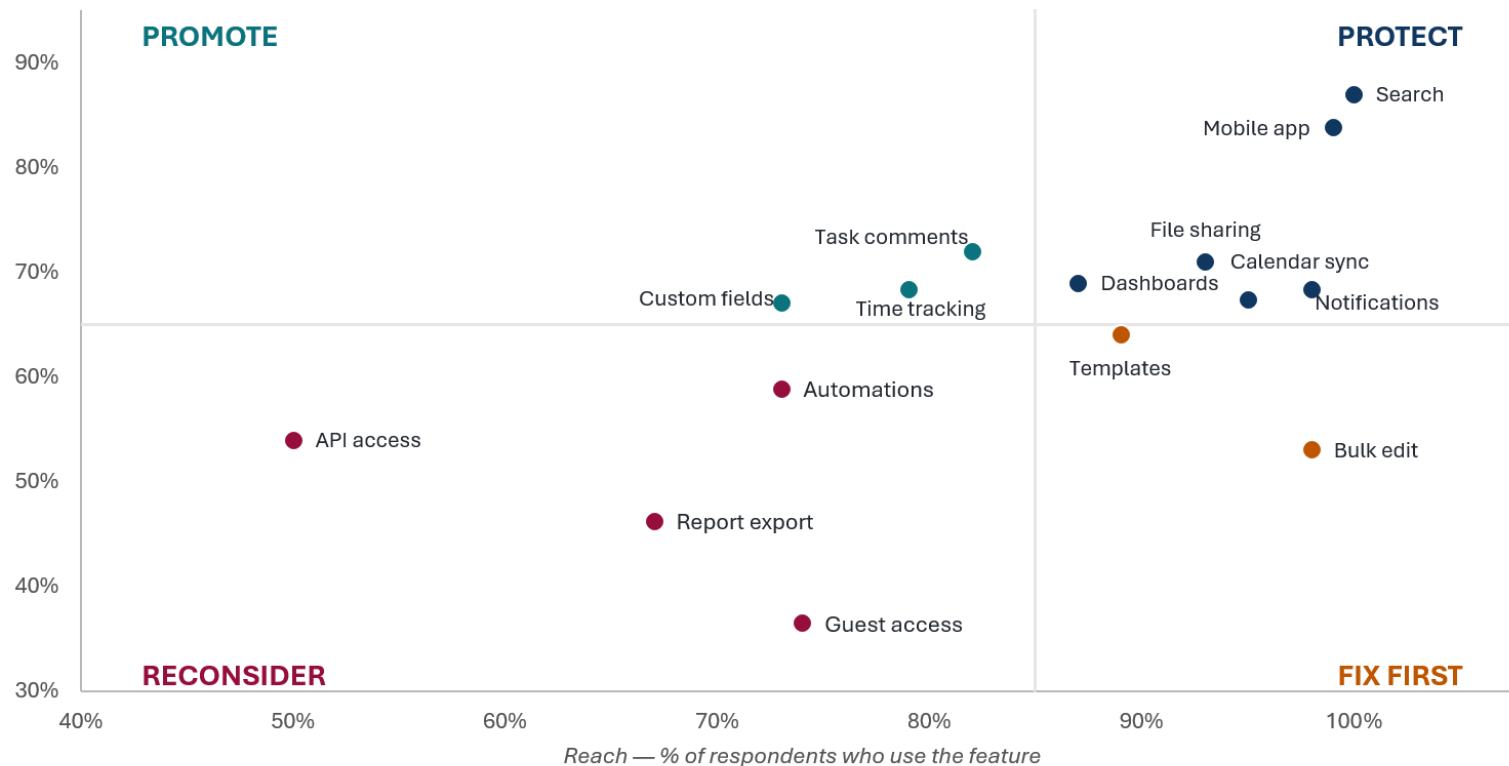
Feature	Have not used	Not satisfied at all	Not very satisfied	Somewhat satisfied	Very satisfied	Completely satisfied	Usage	Satisfied of users	Dissatisfied of users
Search	0%	1%	1%	11%	40%	47%	100%	87%	2%
Mobile app	1%	2%	1%	13%	36%	47%	99%	84%	3%
Notifications	2%	2%	5%	24%	34%	33%	98%	68%	7%
File sharing	7%	2%	4%	21%	37%	29%	93%	71%	6%
Calendar sync	5%	2%	6%	23%	36%	28%	95%	67%	8%
Dashboards	13%	2%	5%	20%	35%	25%	87%	69%	8%
Task comments	18%	3%	5%	15%	26%	33%	82%	72%	10%
Templates	11%	3%	6%	23%	32%	25%	89%	64%	10%
Time tracking	21%	3%	5%	17%	27%	27%	79%	68%	10%
Bulk edit	2%	8%	14%	24%	27%	25%	98%	53%	22%
Custom fields	27%	3%	4%	17%	28%	21%	73%	67%	10%
Automations	27%	3%	4%	23%	27%	16%	73%	59%	10%
Report export	33%	3%	8%	25%	18%	13%	67%	46%	16%
Guest access	26%	9%	14%	24%	17%	10%	74%	36%	31%
API access	50%	2%	6%	15%	16%	11%	50%	54%	16%

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS

TASK COMMENTS and TIME TRACKING are liked but under-adopted

Satisfied share of users (%) against reach (% of respondents using it)



ANOTHER ANGLE

- Reach across, satisfied share of the people who use it up. Two derived numbers replace the six raw ones.
- The quadrants **name the action**: promote, protect, fix first, reconsider.
- Task comments and time tracking are liked but under-adopted — a marketing job, not an engineering one.

The right chart is sometimes for a table you **have not calculated yet**.

TIPS FOR EFFECTIVE VISUALIZATION

CONTEXT MATTERS — DO NOT LET READER GUESS

1 Object — what is actually being measured?

Not "sales". *Units sold, revenue booked, or cash collected?* And whose — the group's, or head office only?

2 Unit — what scale are the numbers in?

VND million · % of respondents · days of inventory · index, 2019 = 100. A number without a unit is a rumour.

3 Time range & category — which period, and which slice?

FY2026, or the twelve months to August 2026? Which branches, which customers — and who was left out.

4 Other context — what makes the number mean something?

The benchmark, the target, last year — or the event that explains the turn: the new WMS, the closure, the lockdown.

TIPS FOR EFFECTIVE VISUALIZATION

1 Choose the right *[basic]* chart

Functionality? Familiarity?

2 Substance over form

Pretty charts is NOT always good charts

3 Declutter, declutter, decluter

Delete until the next delete causes lacking of context

4 Have a message

Communicate an insight! *[but max 2 insights]*

5 Context matters

Enough to understand? Answer to the question?

6 Colors

50 Shades of Grey + ONE [max THREE] colors + shades

TIPS FOR EFFECTIVE VISUALIZATION

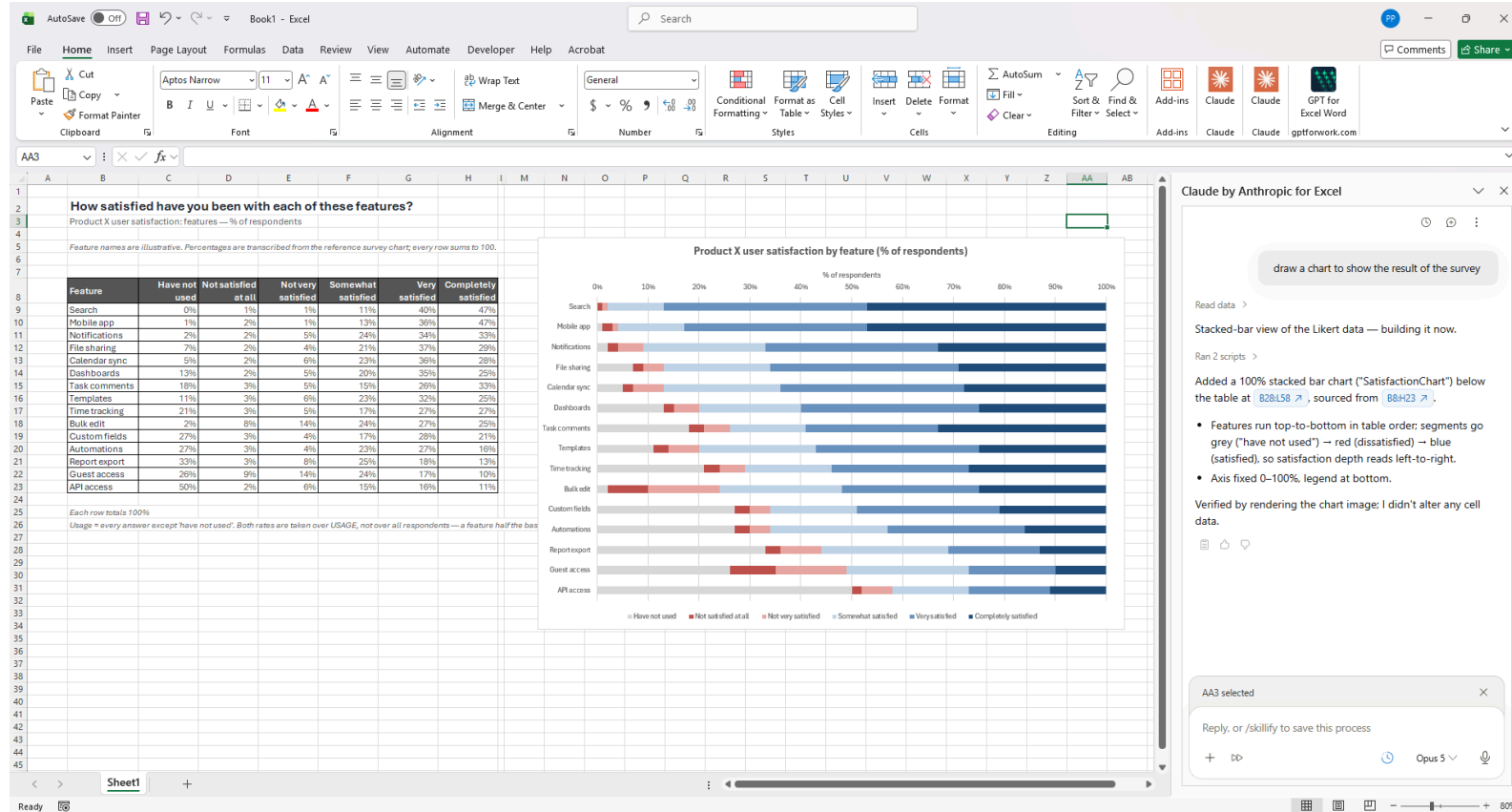
LAY IT OUT

Title <i>States the insight — the one sentence you want remembered</i>	12%
Subtitle	8%
Visual field <i>Axes, gridlines, data series, legend, labels, callouts</i>	75%
Source line	5%

- **Title — 12%.** One sentence carrying the message, in the largest type on the page. Not *"Revenue by branch"*.
- **Subtitle — 8%.** What is measured: the unit, the period, who was counted. Or an additional context/insight.
- **Field — 75%.** The data. **Three quarters of the space belongs to the picture;** everything else is there to serve it.
- **Source line — 5%.** Where the numbers came from, and what is missing. Small, grey, never competing.

AI ADD-IN INSIDE EXCEL

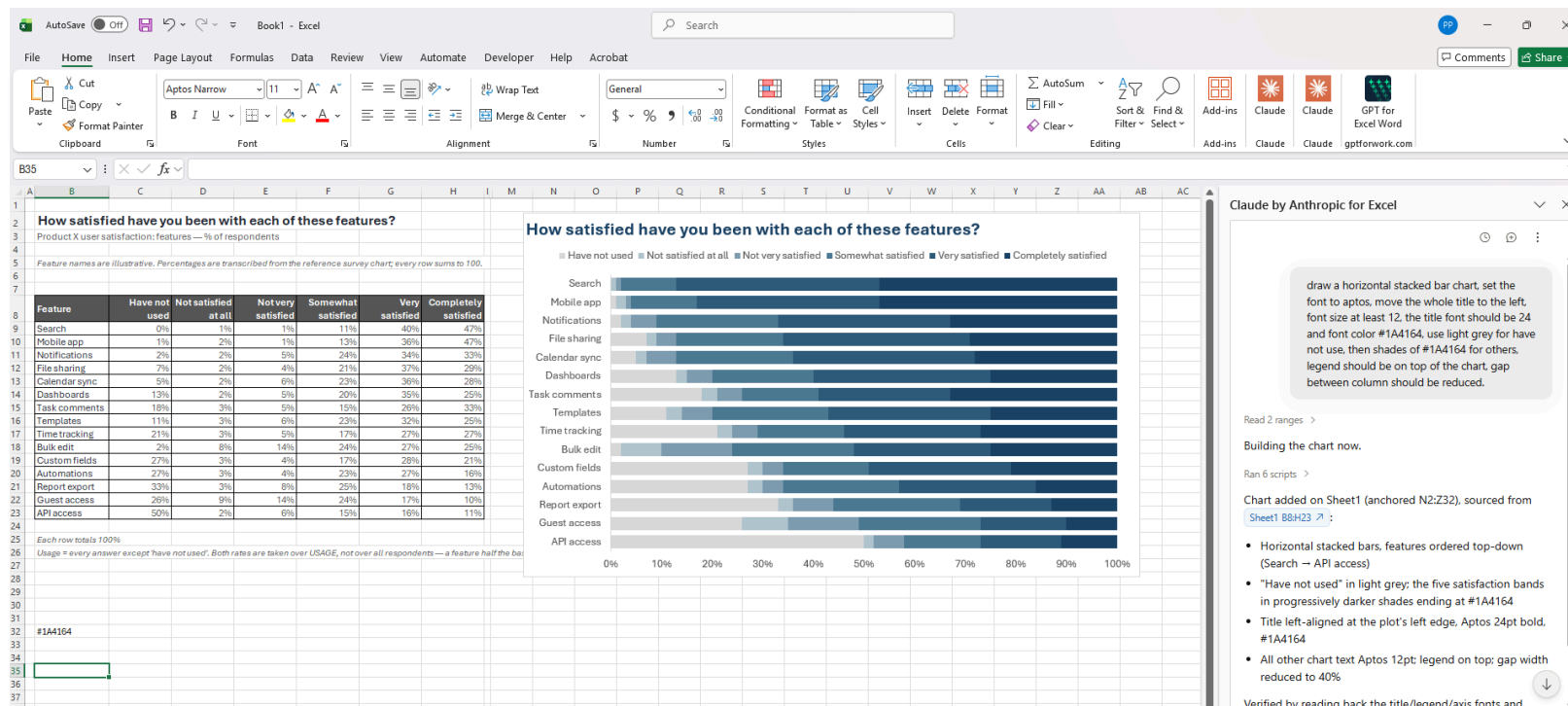
AI IS ALREADY IN THE SPREADSHEET



draw a chart to show the result of the survey

AI ADD-IN INSIDE EXCEL

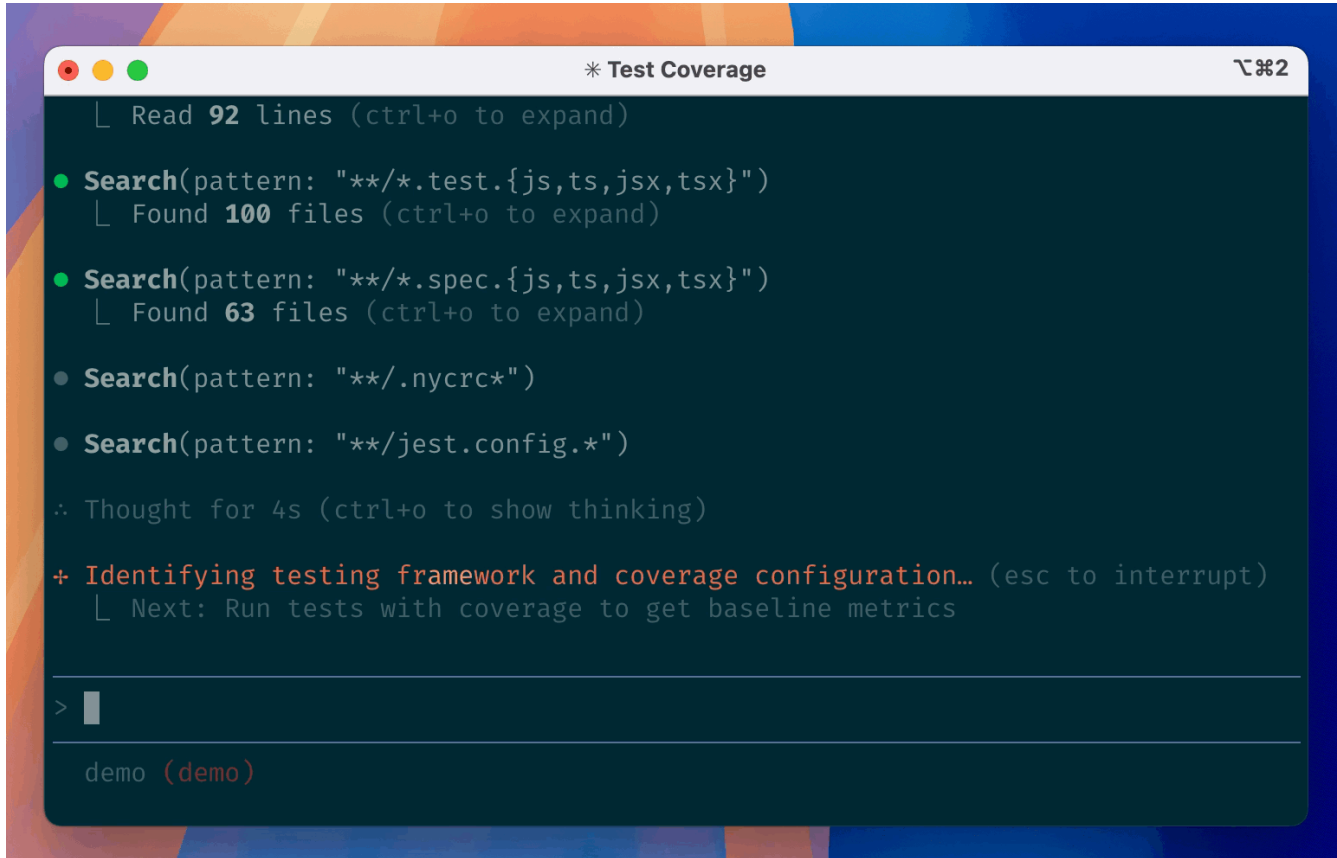
AI IS ALREADY IN THE SPREADSHEET



draw a horizontal stacked bar chart, set the font to aptos, move the whole title to the left, font size at least 12, the title font should be 24 and font color #1A4164, use light grey for have not use, then shades of #1A4164 for others, legend should be on top of the chart, gap between column should be reduced.

AI IN THE TERMINAL

AND IN THE TERMINAL



```
* Test Coverage Ƶ%2
└ Read 92 lines (ctrl+o to expand)
• Search(pattern: "**/*.test.{js,ts,jsx,tsx}")
  └ Found 100 files (ctrl+o to expand)
• Search(pattern: "**/*.spec.{js,ts,jsx,tsx}")
  └ Found 63 files (ctrl+o to expand)
• Search(pattern: "**/*.nycrc*")
• Search(pattern: "**/jest.config.*")
∴ Thought for 4s (ctrl+o to show thinking)
+ Identifying testing framework and coverage configuration... (esc to interrupt)
  └ Next: Run tests with coverage to get baseline metrics

> █
demo (demo)
```

A TERMINAL AGENT

- The chat window **tells you** what to do. An agent in the terminal **does it**: writes the script, runs it, opens Excel, exports the picture — and then looks at what came out.
- That loop is the point. It can **check its own work** and fix it before you ever see it.

ADD-IN OR TERMINAL?

THE SAME MODEL, TWO VERY DIFFERENT SEATS IN THE SPREADSHEET

Sits **inside the file you already have open**: it sees the sheet, the formulas, the named ranges.

- **Interactive.** The chart lands in the workbook. Drag the legend, fix the one stray label, change a colour — the small human edits are a click, not another prompt.
- **It knows the cells.** *"Why is C42 this number?"* — it walks the formula back and cites the cells.
- **Nothing to install.** No terminal, no Python, no repository.
- **You re-describe the style every time.** Each chat starts blank, so the look drifts chart to chart.
- **Bounded by Excel.** One workbook, one chart, whatever the chart engine can draw.

IN THE TERMINAL

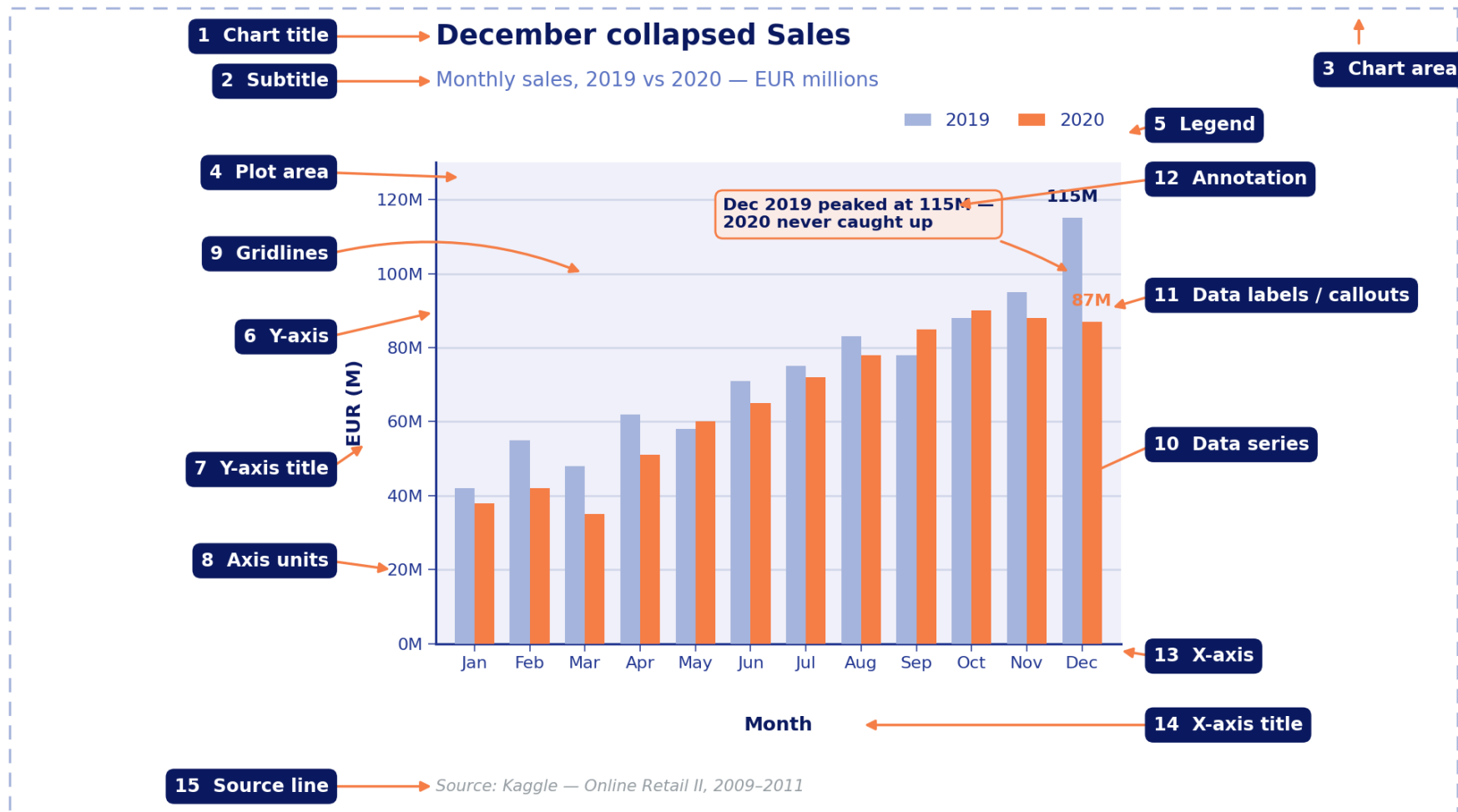
Works **on** the files rather than in them: writes the script, runs it, exports the picture — then looks at what came out.

- **A markdown style guide does the describing.** Write `style_chart.md` once; after that *"draw a bar chart"* is the whole prompt — and what comes back is finished, and identical to the last one.
- **Complex charts live in markdown too.** The rose chart, the small multiples: the recipe written down once, replayed on demand.
- **Python is on the table.** Reshape 200,000 rows, draw what Excel cannot — and still hand back an .xlsx.
- **It repeats and it scales.** Twenty charts, six workbooks, one pass; re-run next month on new data.
- **It cannot touch an open file.** Close the workbook first — and you do not nudge anything while it works.

The add-in is for **the chart in front of you**. The agent is for **the hundred you will build the same way again**.

TIPS FOR EFFECTIVE VISUALIZATION

KNOW THE CHART ELEMENTS



THANK YOU

Do not let the reader scramble for the insight on their own.

QUESTIONS?

CHOOSE THE RIGHT CHART · SUBSTANCE OVER FORM · DECLUTTER · HAVE A MESSAGE

CONTEXT MATTERS · LAY IT OUT · KNOW THE CHART ELEMENTS